Autovacuum Is Running. Is It Actually Working?

adyen

engineered
for ambition

Barna... @barnabasd... · Aug 6, 2025
Honestly the mysql approach sounds better overall.

1 · 6 · 1.4K

Ben Dicken
@BenjDicken

It's nice to never have to think about VACUUM!

5:56 PM · Aug 6, 2025 · **1,299** Views

Theo - t3.gg @theo · Oct 2, 2025
Moving to Postgres was a mistake

281 · 59 · 1K · 874K

John @johnawahba · Oct 2, 2025
Write load got you down?

1 · 15 · 46K

Theo - t3.gg @theo · Oct 2, 2025
Index bloat mostly

17 · 3 · 89 · 50K

The Part of PostgreSQL We Hate the Most

Posted on April 26, 2023

July 26, 2016

Why Uber Engineering Switched from Postgres to MySQL

Evan Klitzke

Nik Samokhvalov @samokhvalov · May 16, 2021
VACUUM sucks...

5 · 2 · 19

The Real Operational Cost of Vacuuming in PostgreSQL

Jonathan Miller • 2026-02-09 • Performance

Ben Dicken
@BenjDicken

The advantage for MySQL here is that the data is kept more compactly. In postgres, you need to do occasional VACUUM maintenance to clean up dead row tuples.

3:55 PM · Jul 30, 2025 · **377** Views

And yet it sucks.

And autovacuum sucks automatically.

P.S. I don't have any problems -- not today.

8:49 PM · May 16, 2021

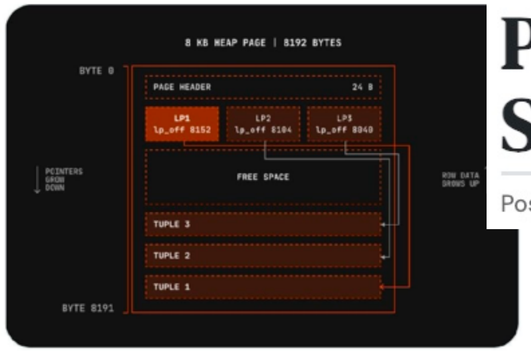
Jeremy Schneider @jer_s · Mar 14, 2024
Replying to @jer_s
🤣🤣 devrim on vacuum: "they asked me to clean my room and I said I'll do it later, because I'm a Postgres person"

1 · 3 · 11 · 774

PlanetScale @PlanetScale · Jul 20
VACUUM is notoriously difficult to get right for busy **Postgres** databases.

The reason it's needed boils down to MVCC and how **Postgres** implements it.

Our latest blog unpacks how it all works.



Dr. Hugo Sánchez
@hochofenbrunza

What's the database internals fact you learned the hard way, in production, at the worst possible time?

I'll start: autovacuum does not care about your deploy window.

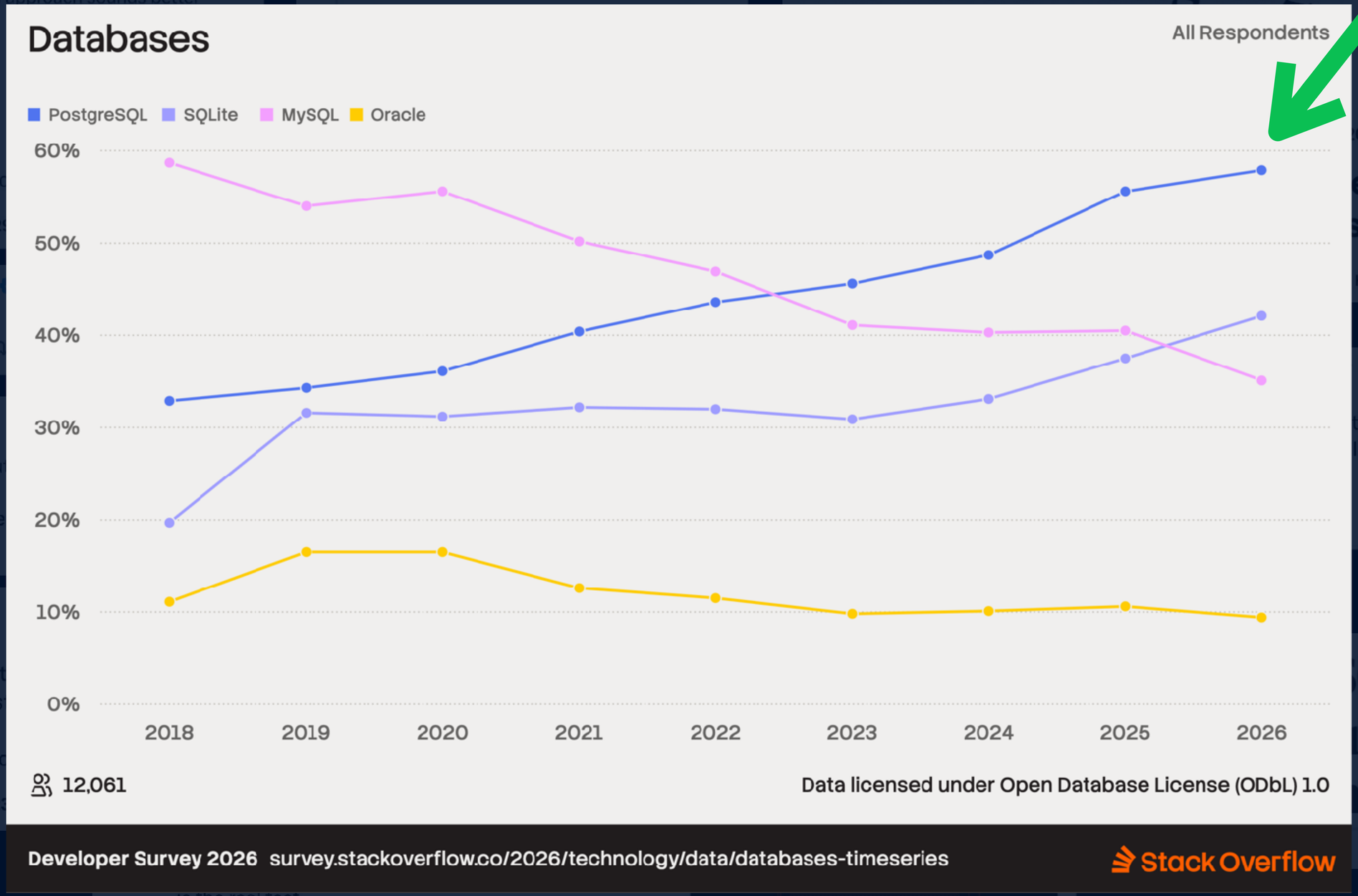
9:22 PM · Sep 22, 2026 · **236** Views

Sujal @hecodesforme · Sep 29
Replying to @wangbin579
Building on Postgres is easy, autovacuum at 3am is the real test

1 · 141

Yes, PostgreSQL Has Problems. But We're Sticking With It!

Posted on June 07, 2023


Databases
All Respondents
PostgreSQL
SQLite
MySQL
Oracle
60%
50%
40%
30%
20%
10%
0%
2018
2019
2020
2021
2022
2023
2024
2025
2026
12,061
Data licensed under Open Database License (ODbL) 1.0
Developer Survey 2026 survey.stackoverflow.co/2026/technology/data/databases-timeseries
Stack Overflow

Vacuum has a PR problem

bug

~~bug~~

lapse

~~bug~~

~~lapse~~

oversight

~~bug~~
~~lapse~~
~~oversight~~

Design
decision

~~bug~~
~~lapse~~
~~oversight~~

Design
decision
Tradeoff

Write now
Cleanup later

Teresa Lopes

Database Engineer

Adyen



Agenda


Why do we need vacuum?


When does autovacuum run?


Is autovacuum keeping up?


When autovacuum cannot run?


What it does not fix?


PostgreSQL 19

Why do we need vacuum?



en.wikipedia.org/wiki/Valencia By Alfred Guesdon

MVCC

Multi-Version Concurrency Control

UPDATE

UPDATE

=

INSERT + DELETE



Dead Tuples



Dead Tuples



Vacuum



Dead Tuples



Autovacuum

(Auto)vacuum phases

- Initializing
- Scanning heap 🔄
- Vacuuming indexes 🔄
- Vacuuming heap 🔄
- Cleaning up indexes
- Truncating heap
- Performing final cleanup

What happens if vacuum does not run?



When does autovacuum run?



What triggers autovacuum?



Anti-wraparound



Anti-
wraparound



Dead
tuples



Anti-
wraparound



Dead
tuples



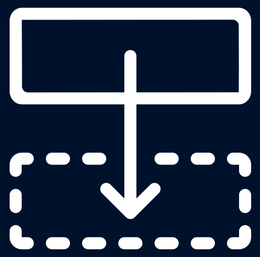
Insert
threshold



Anti-
wraparound



Dead
tuples



Insert
threshold

Our Data Model





10M meals between 1043-1099

every meal **rated 1-5**

and **re-rated** over and over again...

Tables

`el_cid_meals`

10M rows

Table **el_cid_meals**

Column	Type
id	bigint
meal_date	date
restaurante_id	bigint
rating	smallint
note	text

Indexes:
- "el_cid_meals_pkey" PRIMARY KEY, btree (id)
- "el_cid_meals_rating_idx" btree (rating)
- "el_cid_meals_restaurante_idx" btree (restaurante_id)

id	meal_date	restaurante_id	rating
21	1043-01-22	10	2
22	1043-01-23	11	3
23	1043-01-24	12	4
24	1043-01-25	1	5
25	1043-01-26	2	1

...

Tables

`el_cid_meals`

10M rows

`restaurantes`

12 rows

Table **restaurantes**

Column	Type
id	bigint
name	text
since	integer
note	text

Indexes:
    "restaurantes_pkey" PRIMARY KEY, btree (id)

id	name	since	note
1	El Campeador	1094	flagship, table by the window
2	Babieca	1099	named after the horse
3	Dona Jimena	1094	the boss runs it
4	Vivar	1081	first kitchen, in exile
5	La Taifa	1094	acquired from the previous management
6	Albufera	1094	where the rice grows
7	Malvarrosa	1095	beach kiosk
...			
11	El Carmen	1097	old quarter branch
12	Xerea	1098	behind the cathedral

Tables

el_cid_meals

10M rows

restaurantes

12 rows

paella

1 rows

paella_ingredients

12 rows

Table **paella**

Column	Type
id	bigint
name	text
notes	text

Indexes:
"paella_pkey" PRIMARY KEY, btree (id)

Table **paella_ingredients**

Column	Type
id	bigint
name	text
quantity	numeric(6,2)
unit	text

Indexes:
"paella_ingredients_pkey" PRIMARY KEY, btree (id)

id	name	notes
1	Paella Valenciana	the OG one

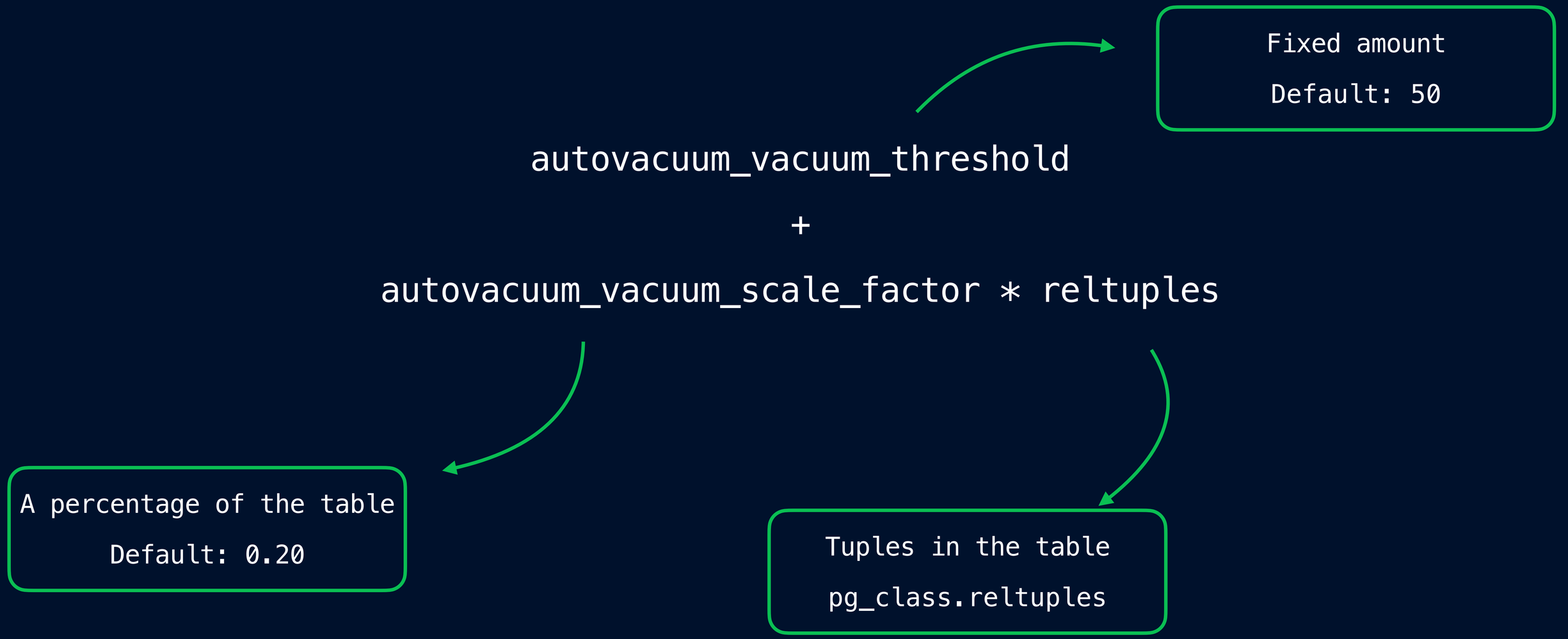
id	name	quantity	unit
1	arroz	320.00	g
2	pollo	800.00	g
3	conejo	600.00	g
4	ferraura	200.00	g
5	garrofo	100.00	g
6	tomate	150.00	g
7	aceite	100.00	ml
8	azafran	0.10	g
9	sal	8.00	g
10	agua	1200.00	ml
11	romero	1.00	branca
12	chorizo	0.00	g







The trigger: when vacuum starts



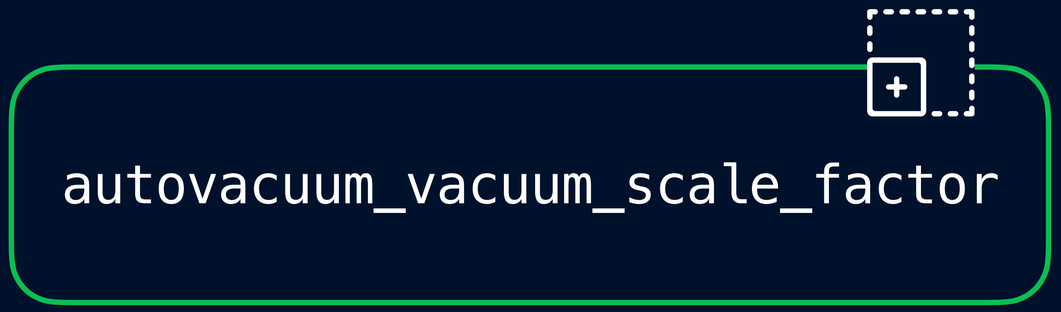
50

+

0.20 * 10 000 000

=

2 000 050 **dead tuples**

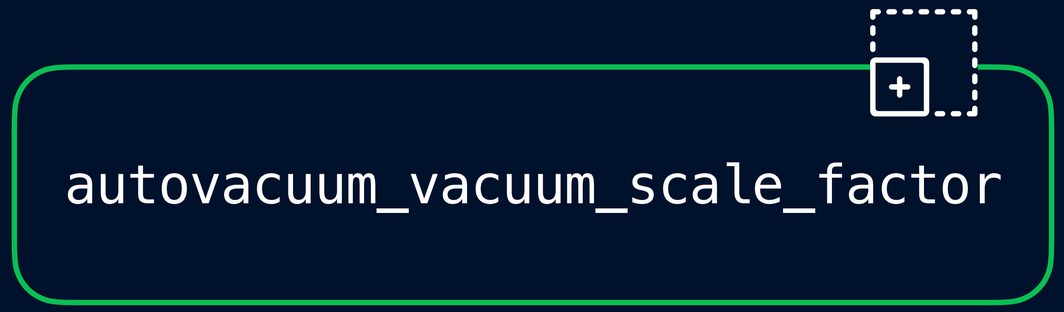


el_cid_meals
10M rows

UPDATE 50 000
rows batches

scale factor	Vacuum trigger	Time*
0.20	2 000 050	49.3 s

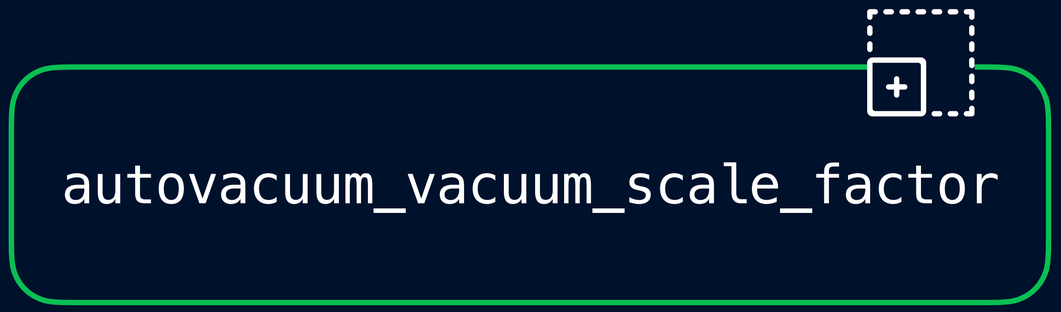
*2vCPUS, 8GB RAM - absolute times are not the most important, but rather the difference between executions



el_cid_meals
10M rows

UPDATE 50 000
rows batches

```
ALTER TABLE el_cid_meals
  SET (autovacuum_vacuum_scale_factor = 0.01);
```



el_cid_meals
10M rows

UPDATE 50 000
rows batches

scale factor	Vacuum trigger	Time*
0.20	2 000 050	49.3 s
0.01	100 050	1.9 s

*2vCPUS, 8GB RAM - absolute times are not the most important, but rather the difference between executions



Big tables, default 0.2 - a huge backlog builds before vacuum starts



`autovacuum_vacuum_threshold`

restaurantes
12 rows

UPDATE
60 dead tuples

Threshold | scale factor	Vacuum trigger	Result
500 | 0.20	~502	0 runs (vacuum not triggered)



`autovacuum_vacuum_threshold`

`restaurantes`
`12 rows`

`UPDATE`
`60 dead`
`tuples`

Threshold | scale factor	Vacuum trigger	Result
500 | 0.20	~502	0 runs (vacuum not triggered)
50 | 0.20	~52	1 run (`tuples: 60 removed`)



`autovacuum_vacuum_threshold`

restaurantes
12 rows

UPDATE
60 dead tuples

Threshold | scale factor	Vacuum trigger	Result
500 | 0.20	~502	0 runs (vacuum not triggered)
50 | 0.20	~52	1 run (`tuples: 60 removed`)
50 | 0	50	1 run (`tuples: 60 removed`)



`autovacuum_vacuum_threshold`

el_cid_meals
10M rows

UPDATE
100 000 dead tuples

Threshold | scale factor	Vacuum trigger	Result
50 | 0.20	2 000 050	0 runs (vacuum not triggered)



`autovacuum_vacuum_threshold`

el_cid_meals
10M rows

UPDATE
100 000 dead tuples

Threshold | scale factor	Vacuum trigger	Result
50 | 0.20	2 000 050	0 runs (vacuum not triggered)
0 | 0.20	2 000 000	0 runs (vacuum not triggered)

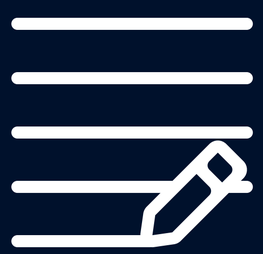


`autovacuum_vacuum_threshold`

`el_cid_meals`
`10M rows`

`UPDATE`
`100 000 dead tuples`

Threshold | scale factor	Vacuum trigger	Result
50 | 0.20	2 000 050	0 runs (vacuum not triggered)
0 | 0.20	2 000 000	0 runs (vacuum not triggered)
50 | 0.005	50 050	1 run (`tuples: 100 000 removed`)



Threshold -> small tables

Scale factor -> big tables

```
autovacuum_vacuum_scale_factor = 0

  autovacuum_vacuum_threshold = 0
```

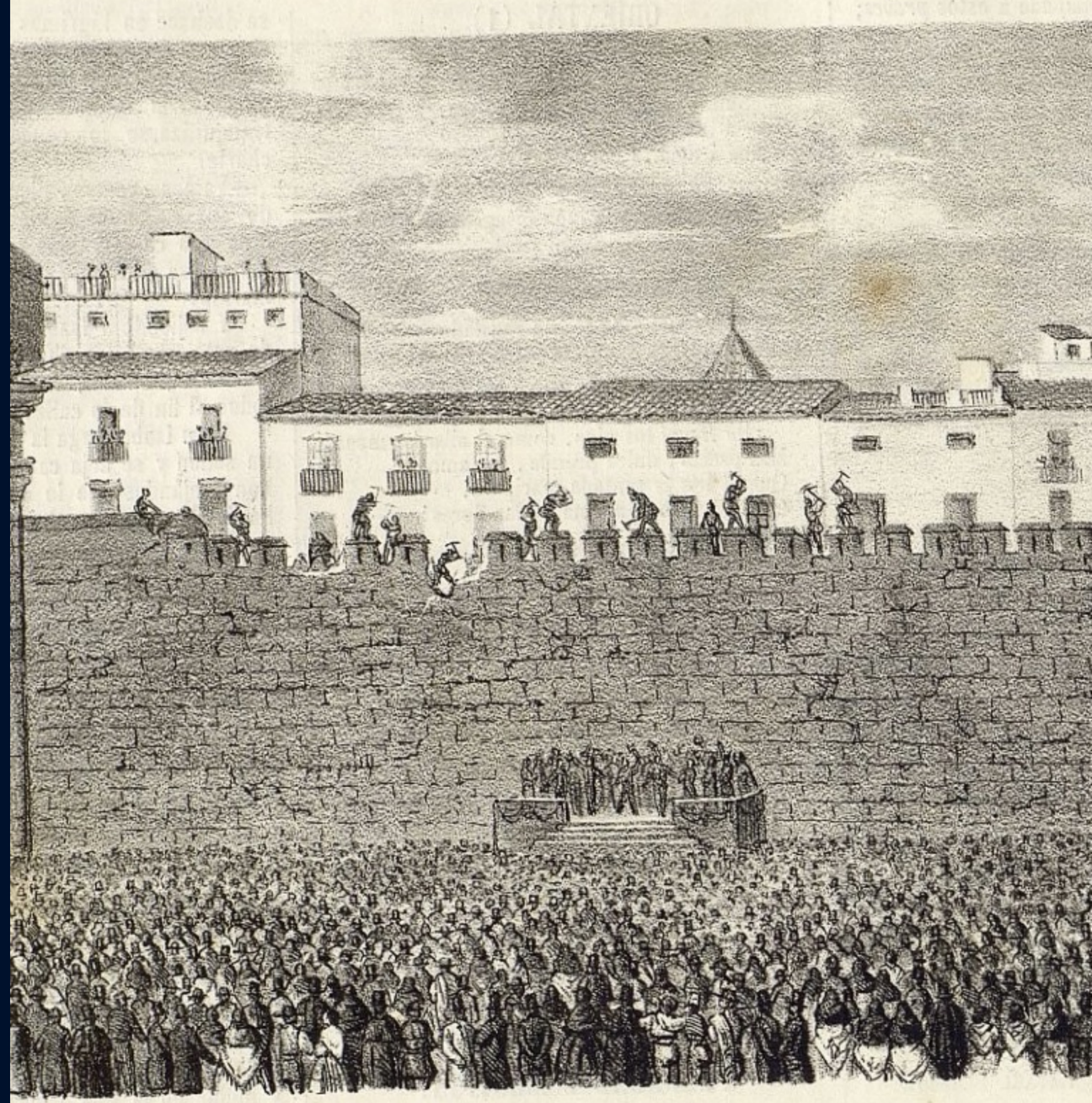
```
autovacuum_vacuum_scale_factor = 0

autovacuum_vacuum_threshold = 0
```



Bad idea, don’t do it!

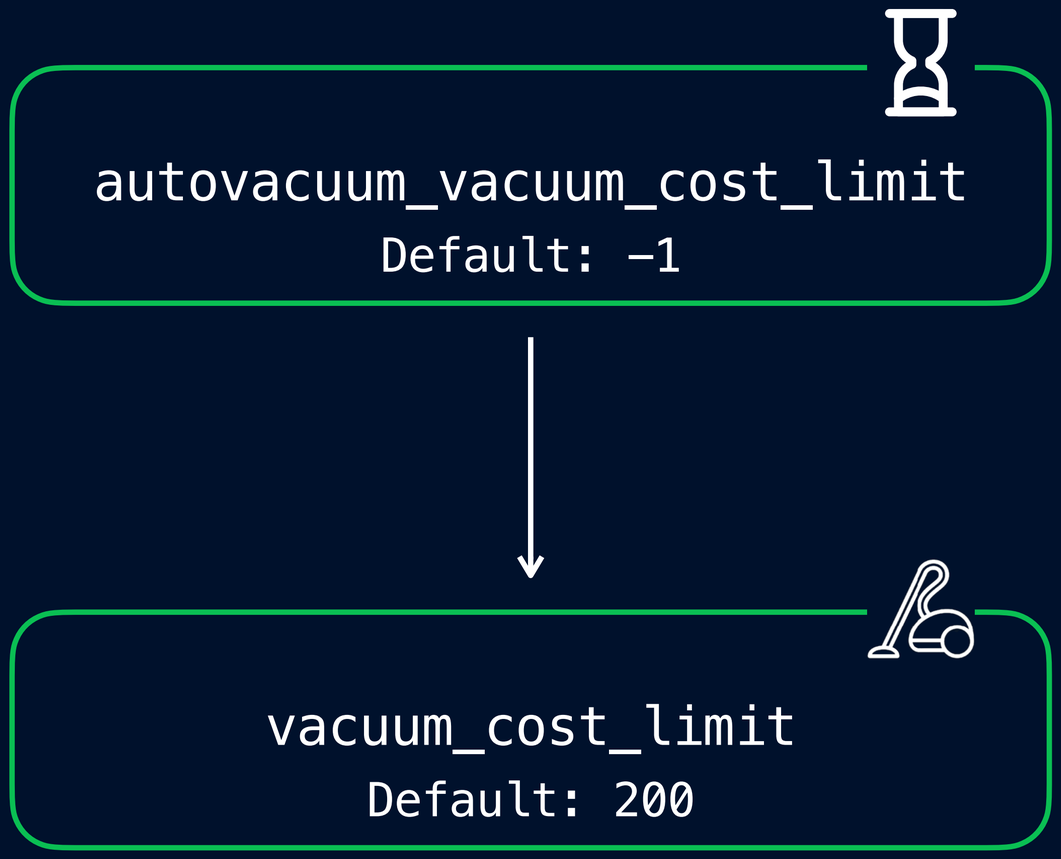
Is autovacuum keeping up?

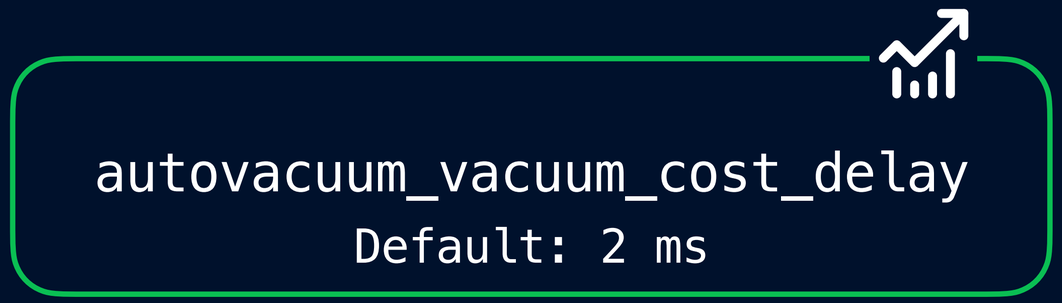


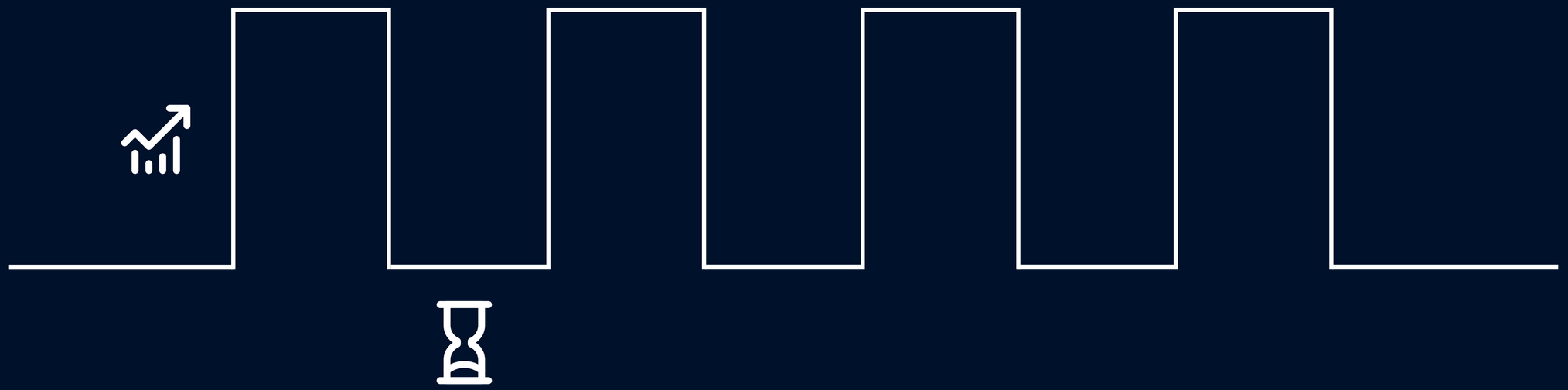
INAUGURACION DE LOS TRABAJOS DEL DERRIBO DE LAS MURALLAS EN VALENCIA
(20 de Febrero de 1865, véase la revista de la semana).
en.wikipedia.org/wiki/Valencia By A. Bergón


cost_delay
cost_limit
imgflip.com
JAKE-CLARK.TUMBLR

Cost limit and cost delay: a delicate balance







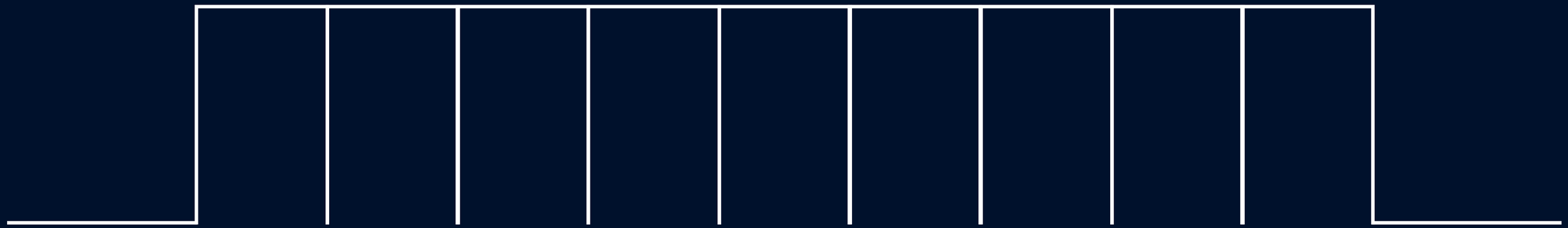
Small cost, big delay



Small cost, big delay



Big cost, small delay

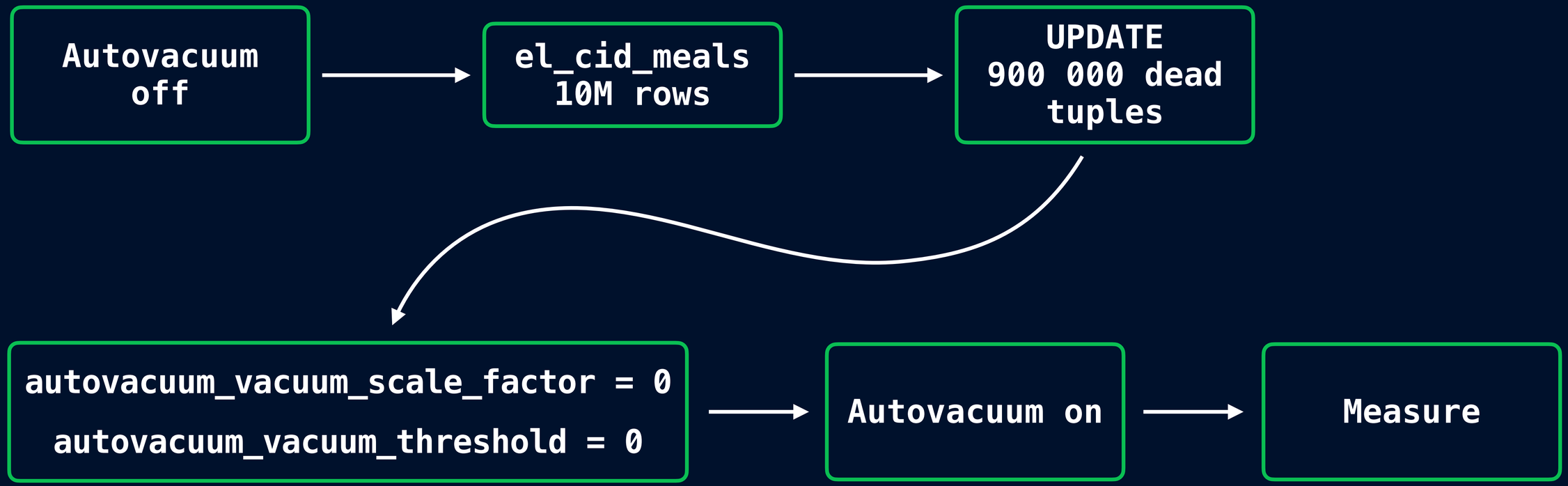


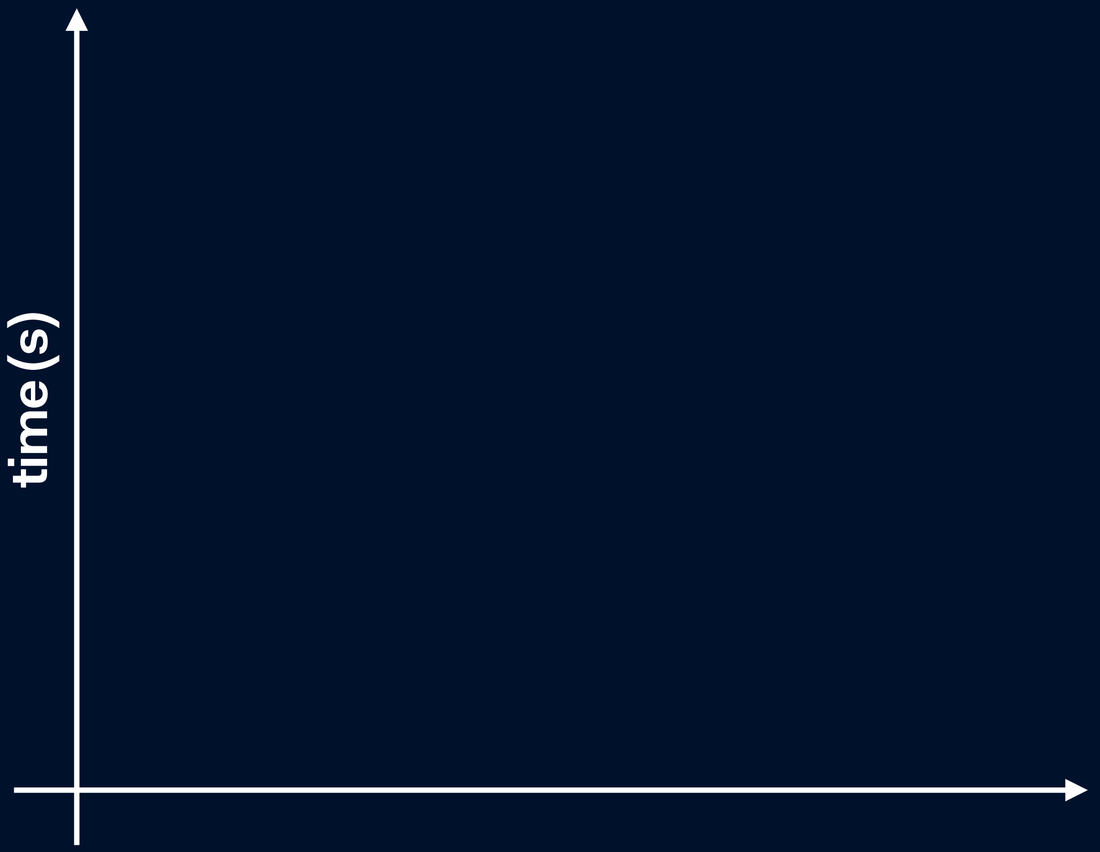
Same backlog, three speeds: the pauses change, the work doesn’t

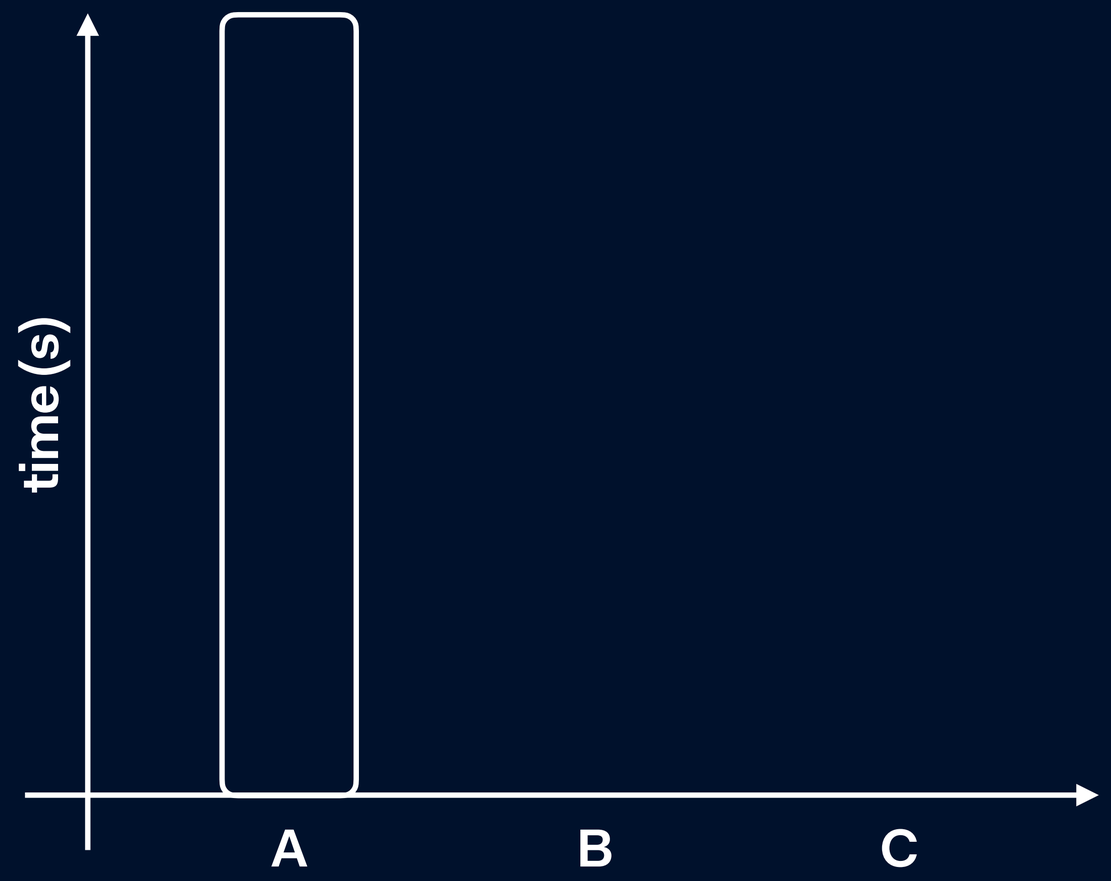
`el_cid_meals`
`10M rows`

	cost_limit	cost_delay
A	200	2 ms
B	2000	2 ms
C	2000	20 ms

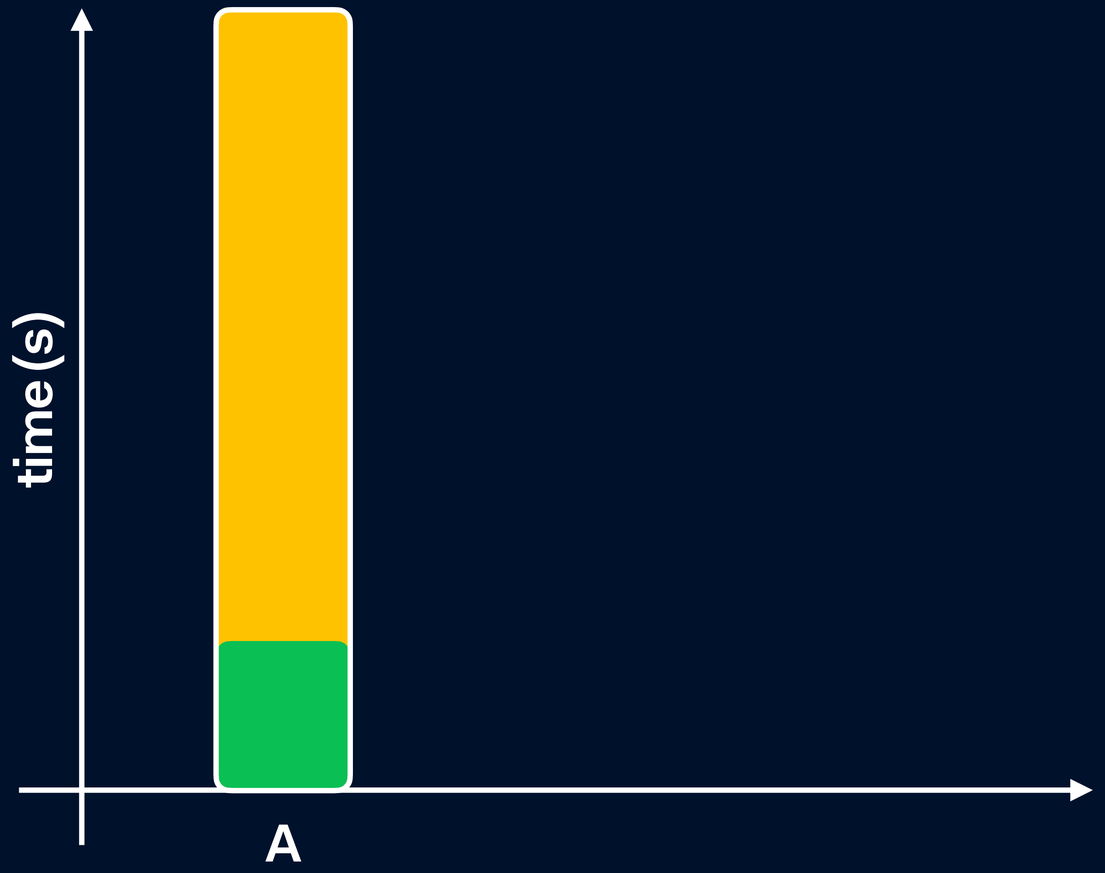
Test setup - *NOT* advice




time (s)



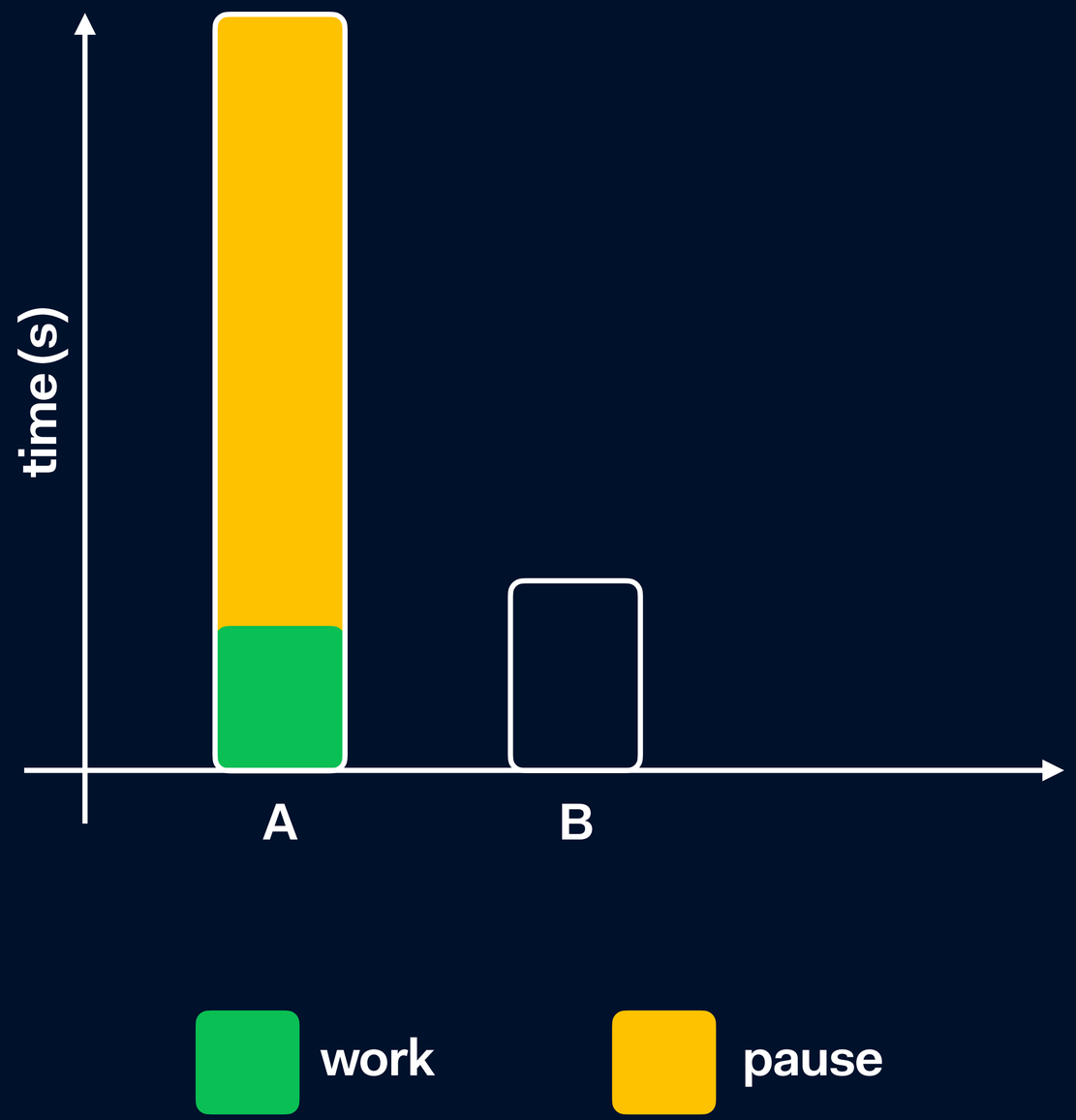
	cost_limit	cost_delay
A	200	2 ms



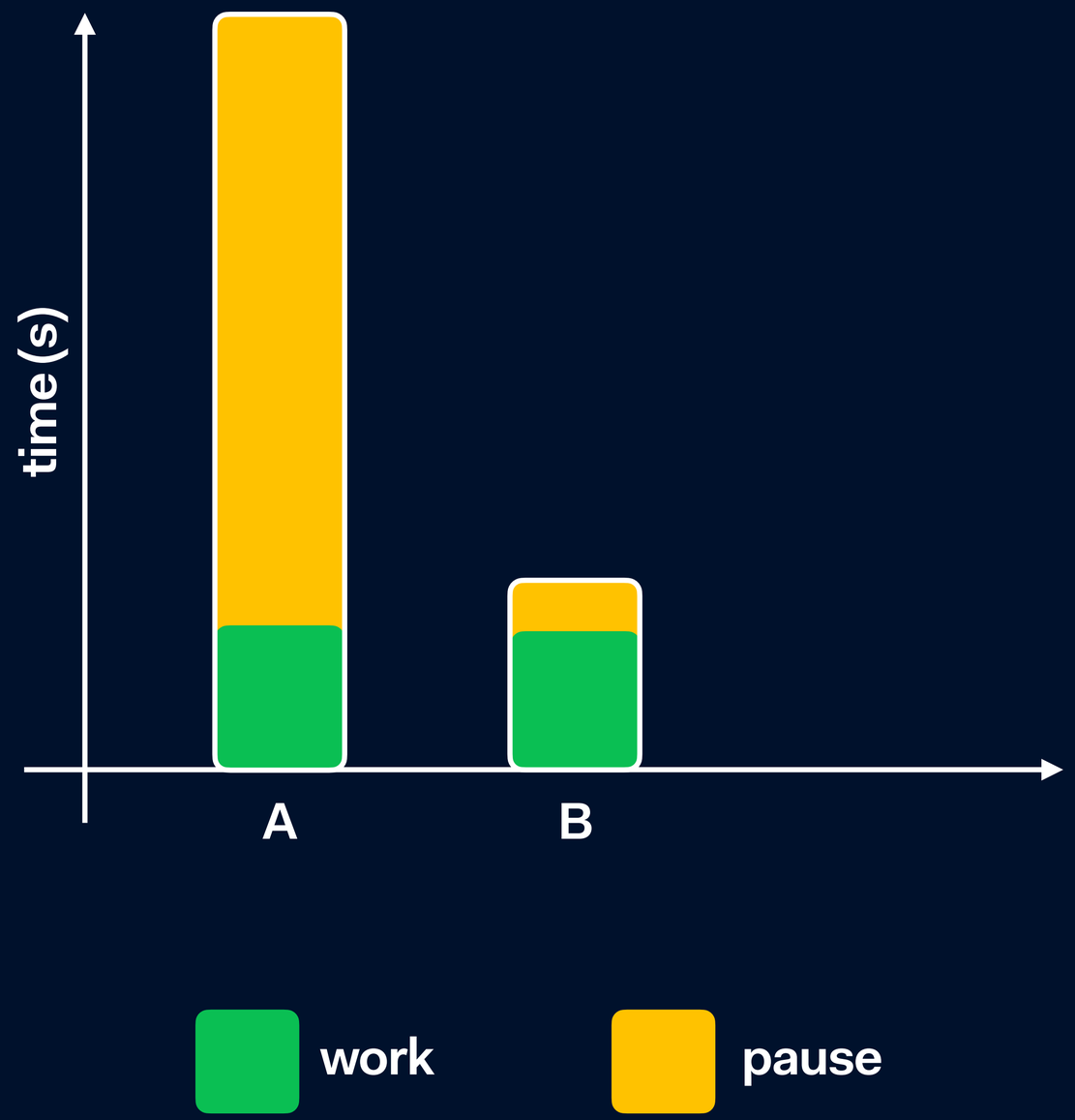
	cost_limit	cost_delay
A	200	2 ms

work

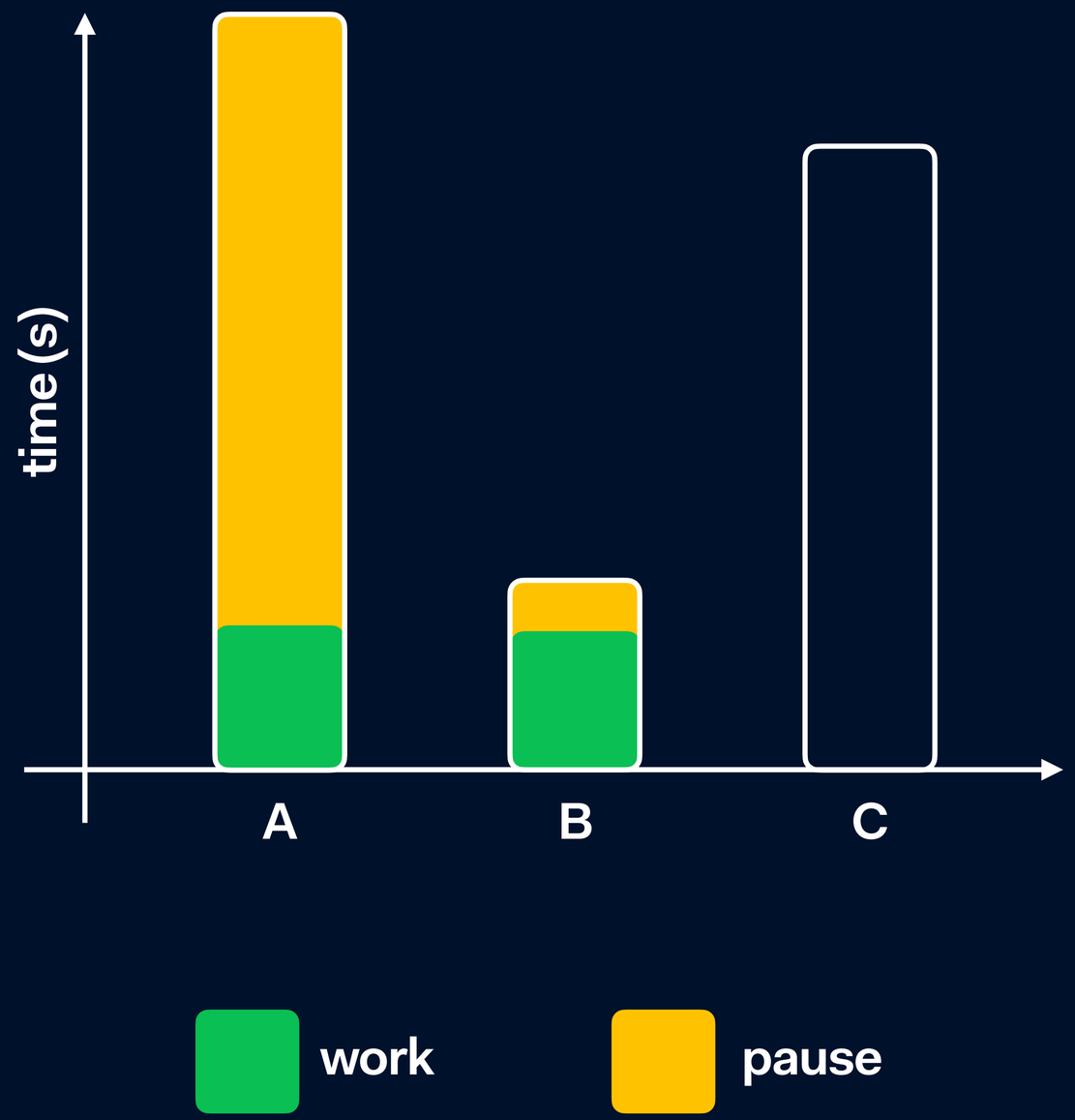
pause



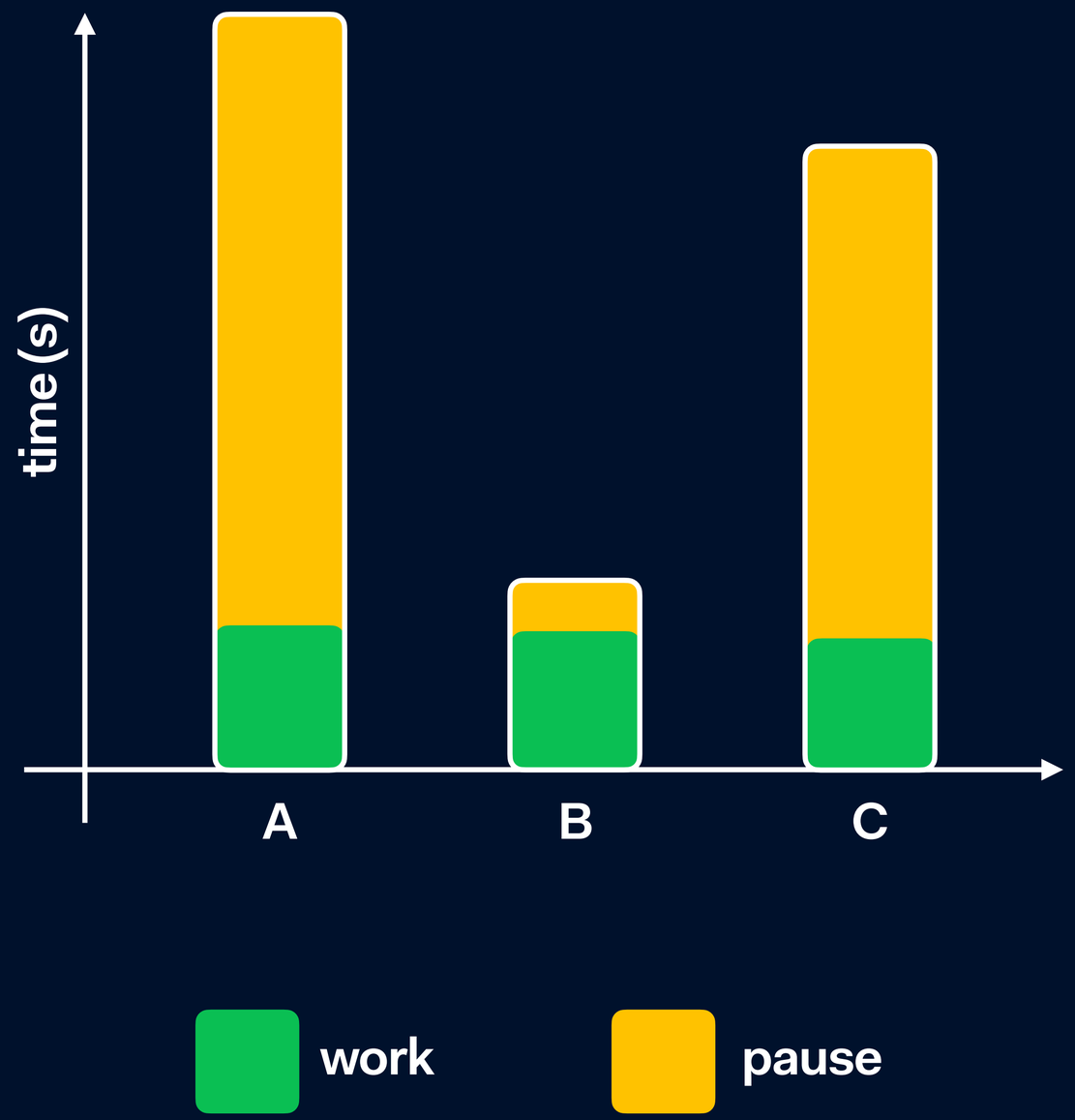
	cost_limit	cost_delay
A	200	2 ms
B	2000	2 ms



	cost_limit	cost_delay
A	200	2 ms
B	2000	2 ms



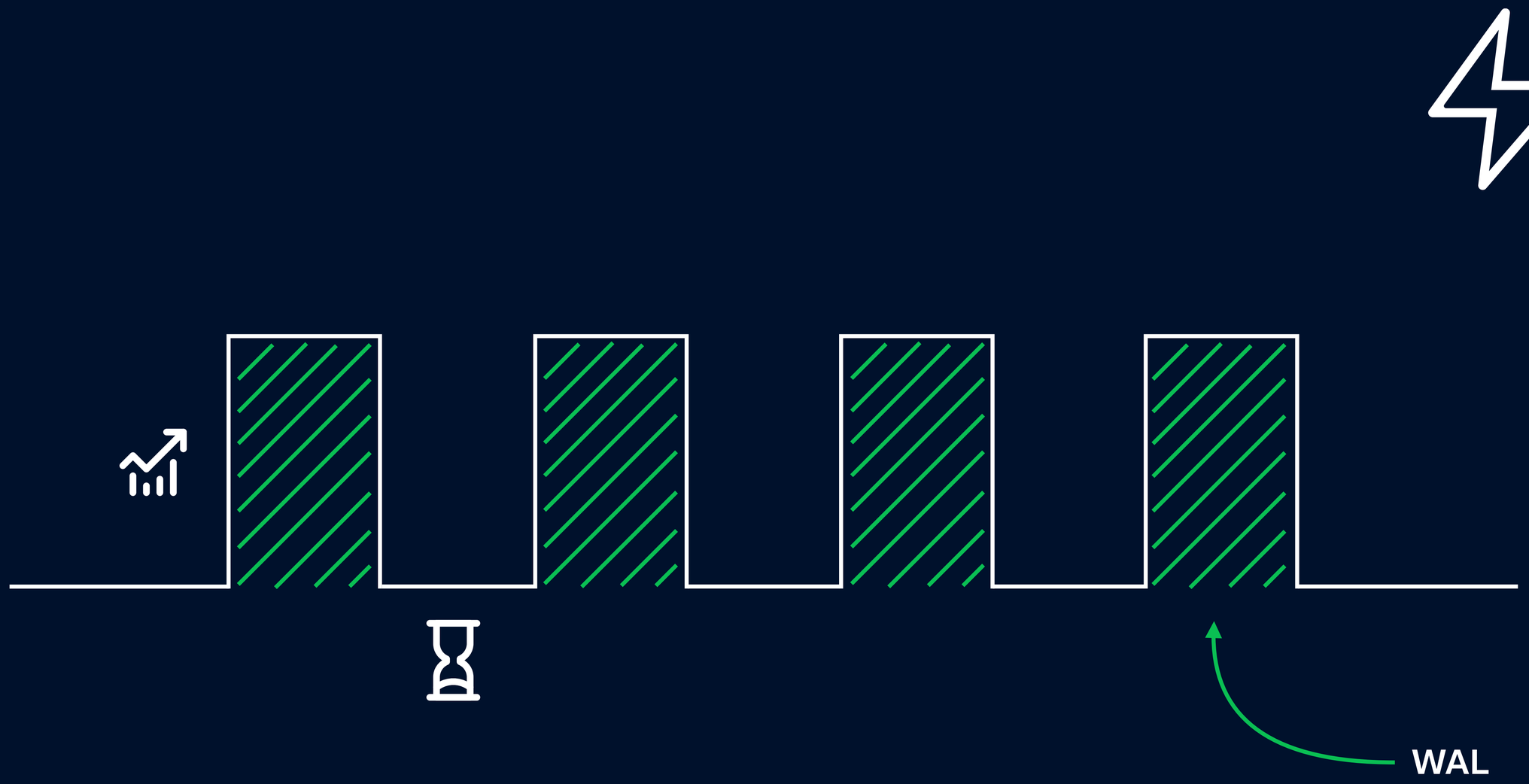
	cost_limit	cost_delay
A	200	2 ms
B	2000	2 ms
C	2000	20 ms



	cost_limit	cost_delay
A	200	2 ms
B	2000	2 ms
C	2000	20 ms



Allow vacuum more work in each run - consider increase cost_limit


WAL



Do not assume - test and measure

Bring all of them together



autovacuum_vacuum_scale_factor



autovacuum_vacuum_threshold



autovacuum_vacuum_cost_limit



autovacuum_vacuum_cost_delay

el_cid_meals
10M rows

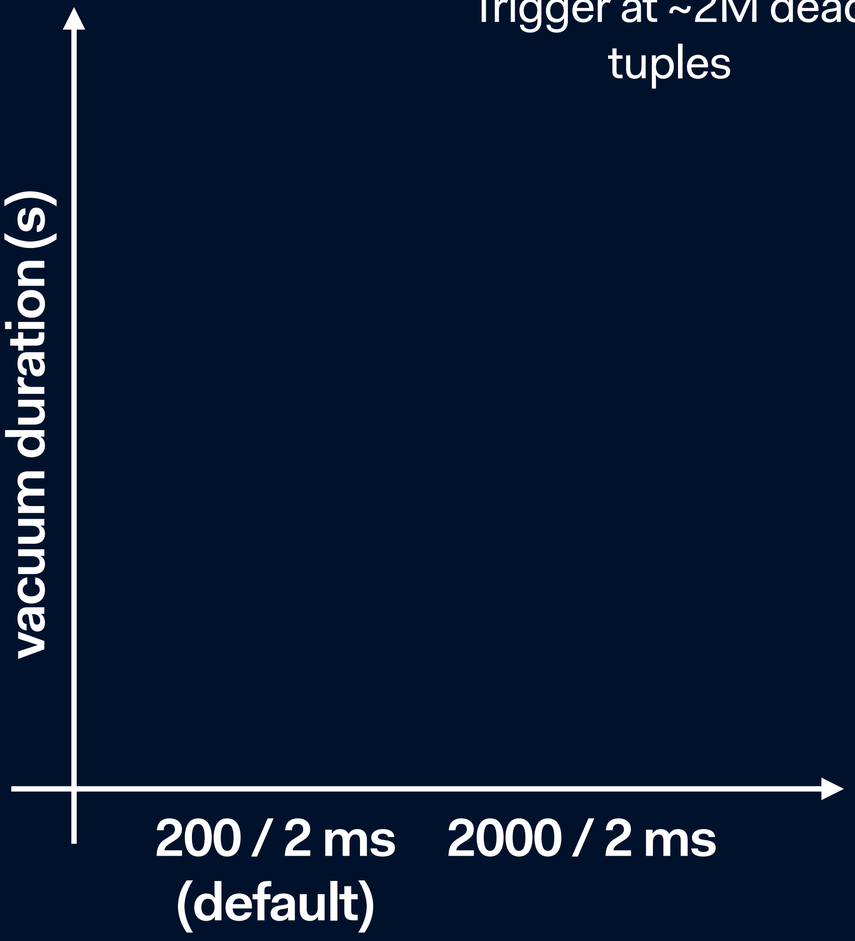
UPDATE 50 000
rows batches

`el_cid_meals`
`10M rows`

`UPDATE 50 000`
`rows batches`

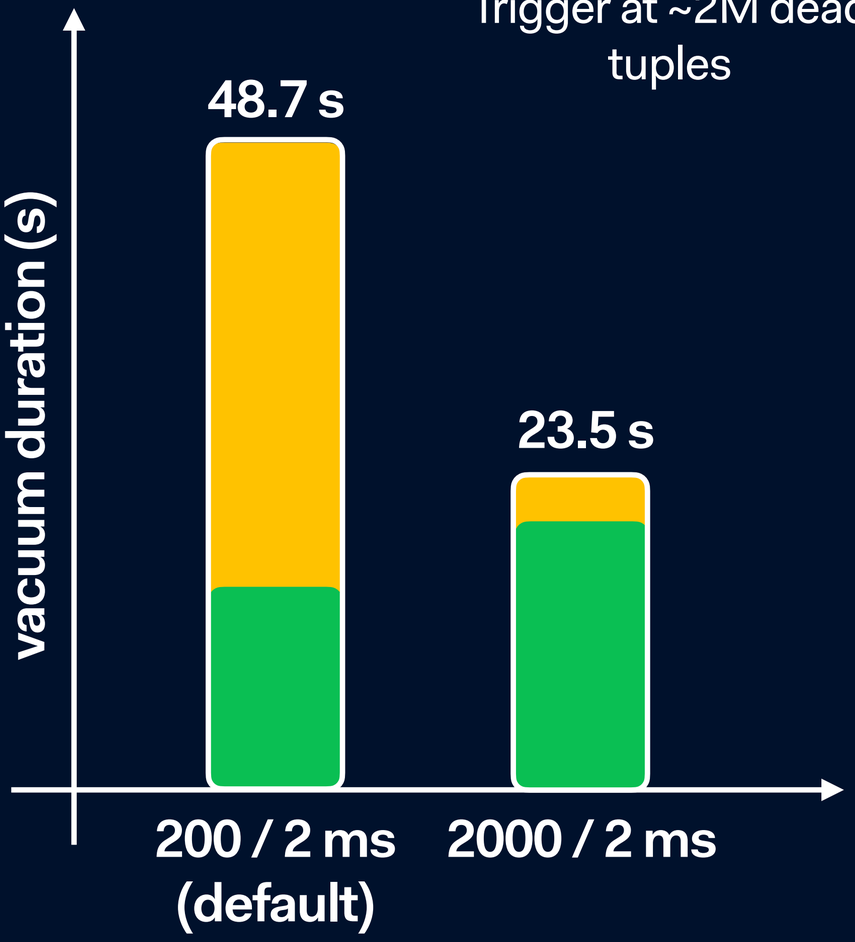
scale_factor = 0.20

Trigger at ~2M dead tuples



scale_factor = 0.20

Trigger at ~2M dead tuples

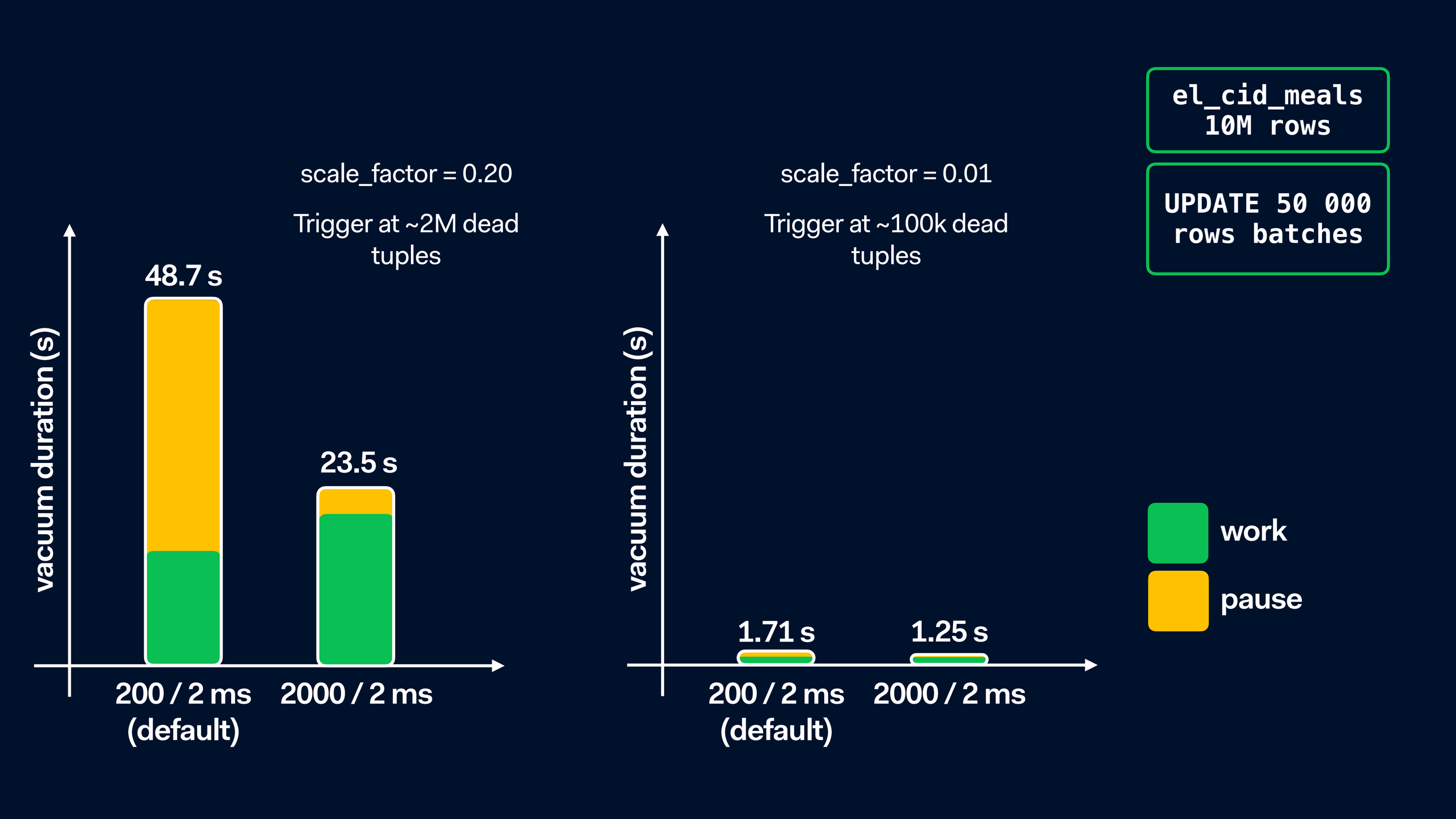


`el_cid_meals`
`10M rows`

`UPDATE 50 000`
`rows batches`

work

pause


el_cid_meals
10M rows
UPDATE 50 000
rows batches
scale_factor = 0.20
Trigger at ~2M dead
tuples
48.7 s
23.5 s
vacuum duration (s)
200 / 2 ms
(default)
2000 / 2 ms
scale_factor = 0.01
Trigger at ~100k dead
tuples
1.71 s
1.25 s
vacuum duration (s)
200 / 2 ms
(default)
2000 / 2 ms
work
pause

1 table 



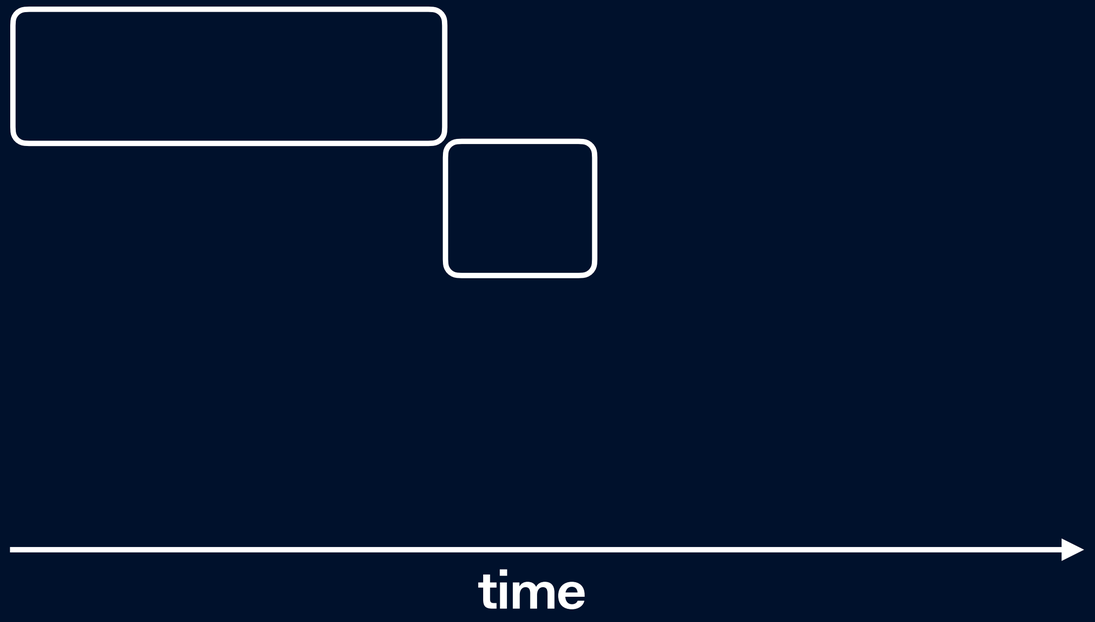
```
autovacuum_max_workers = 1
```





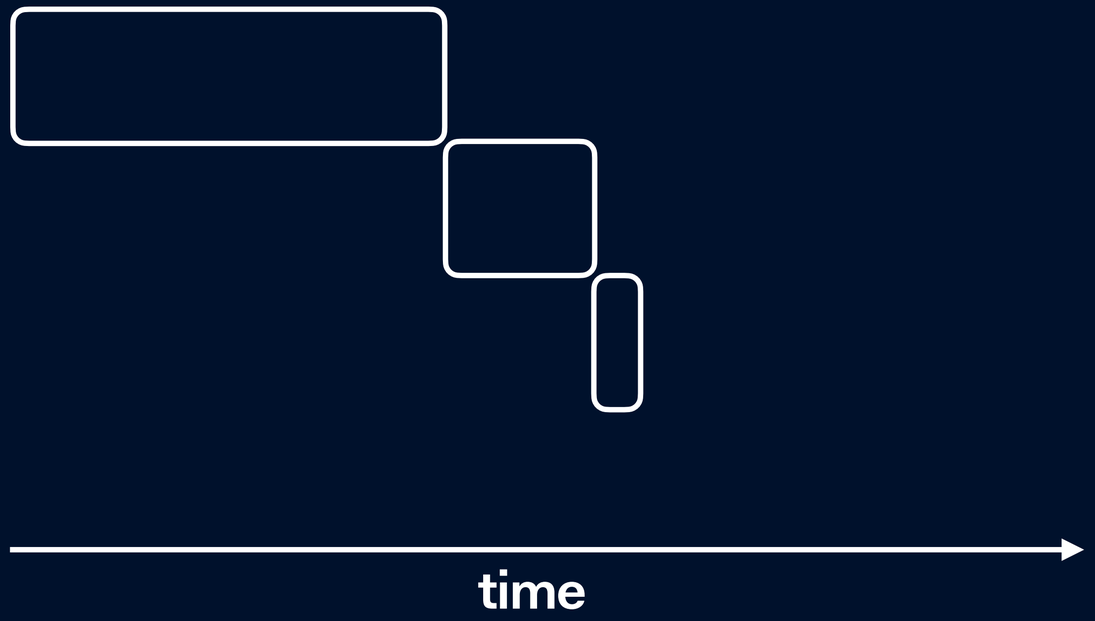


```
autovacuum_max_workers = 1
```



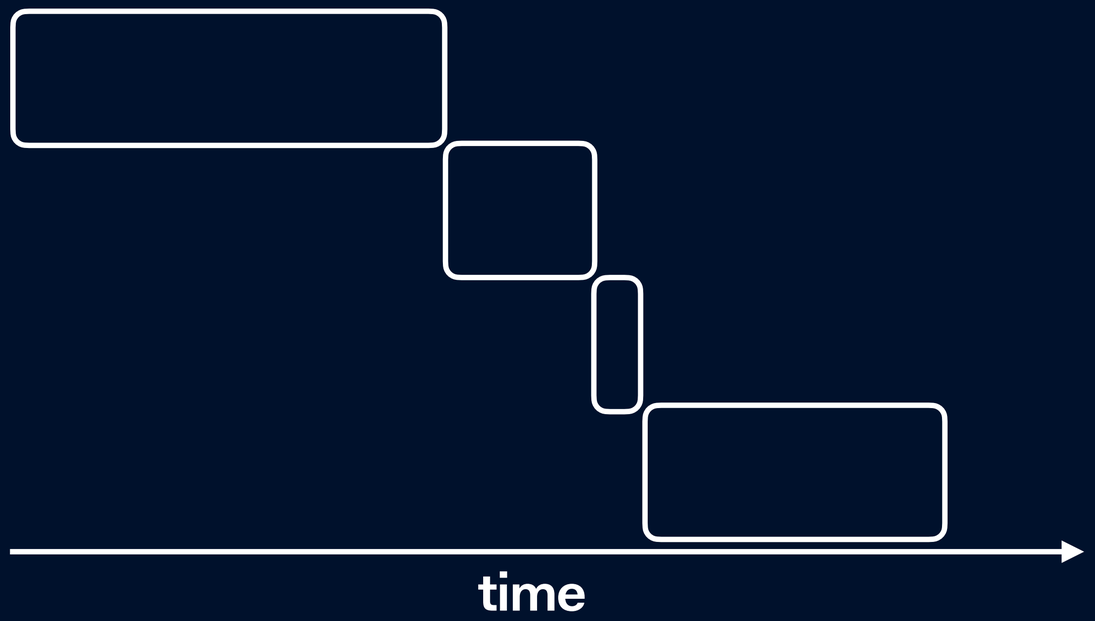


```
autovacuum_max_workers = 1
```

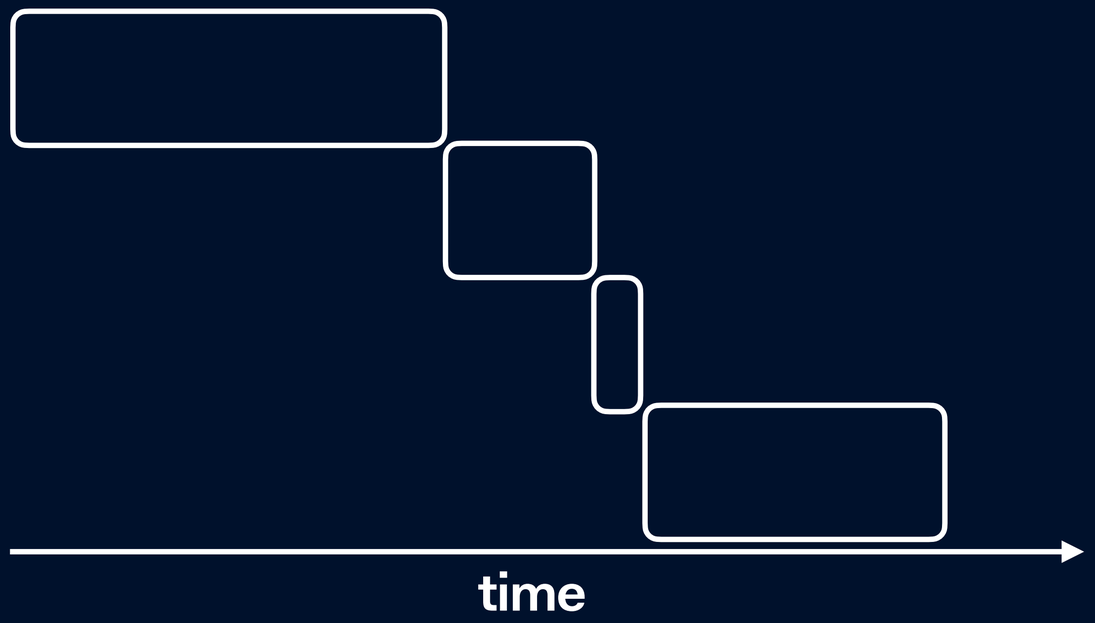




```
autovacuum_max_workers = 1
```



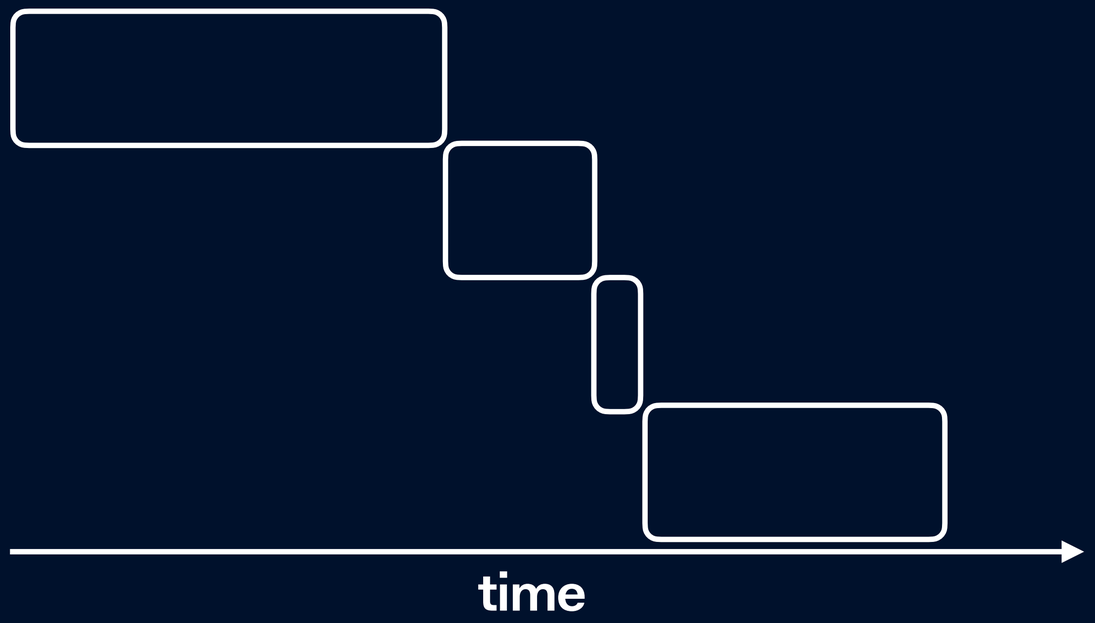
```
autovacuum_max_workers = 1
```



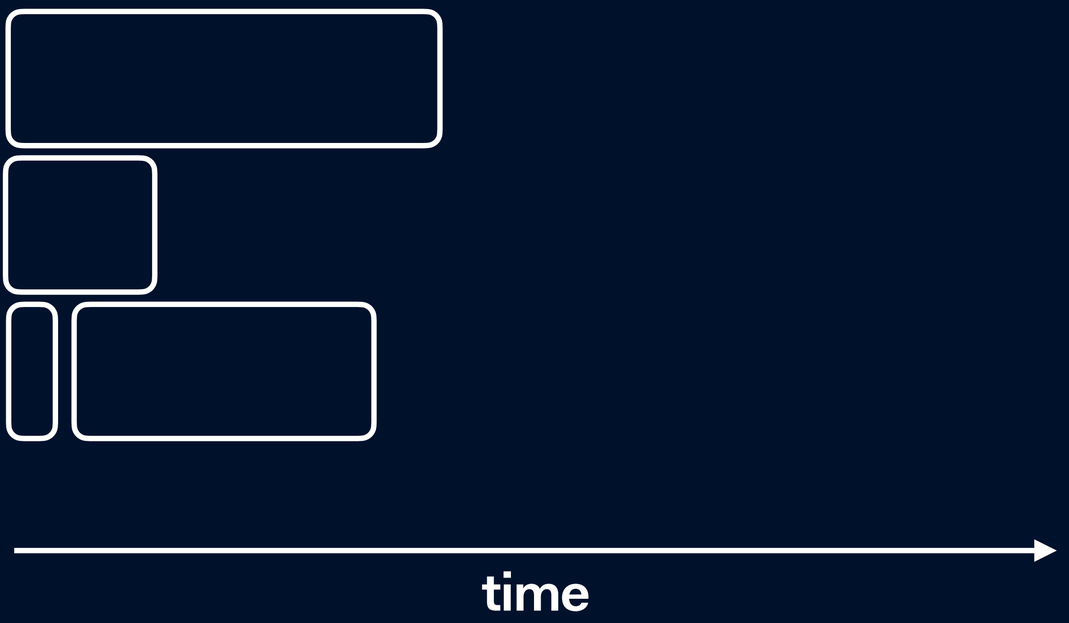

```
autovacuum_max_workers = 3
```

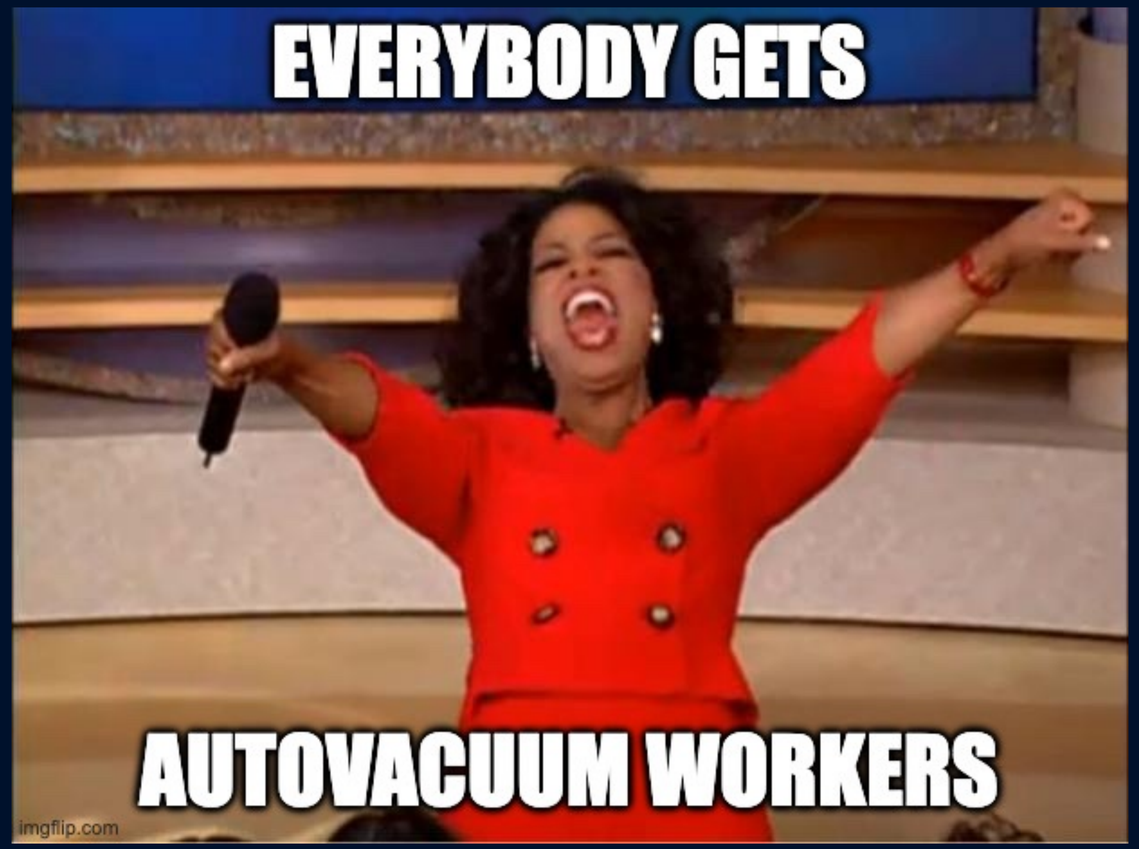


```
autovacuum_max_workers = 1
```




```
autovacuum_max_workers = 3
```




EVERYBODY GETS
AUTOVACUUM WORKERS
imgflip.com

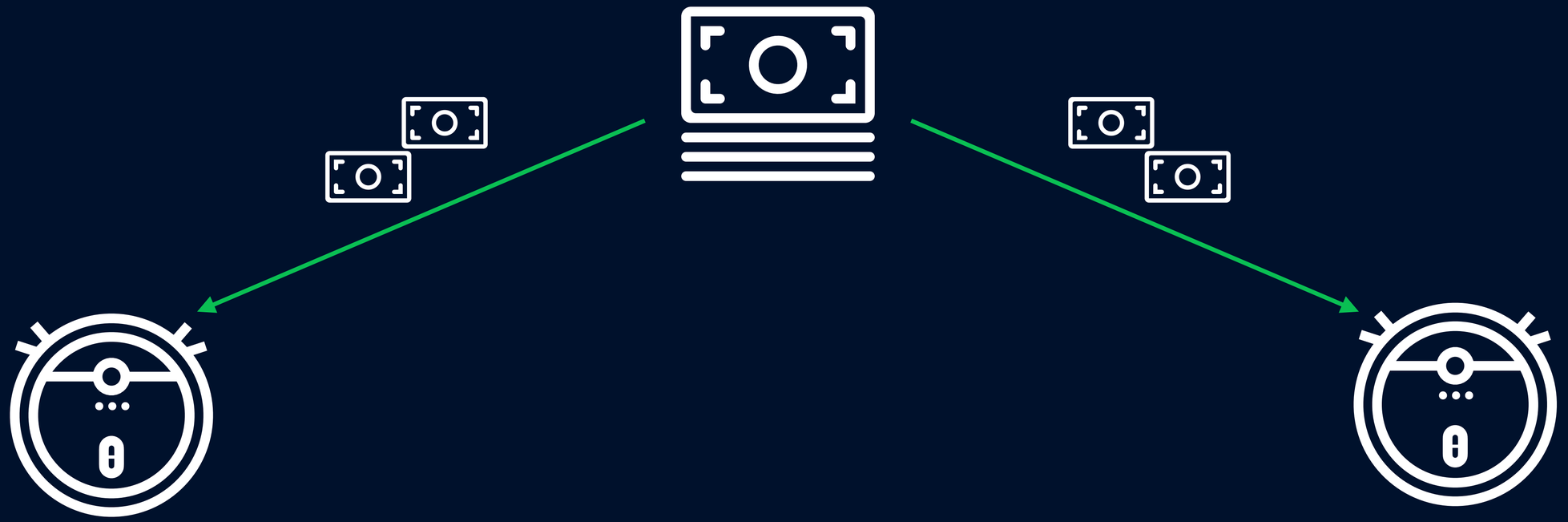
Cost is shared



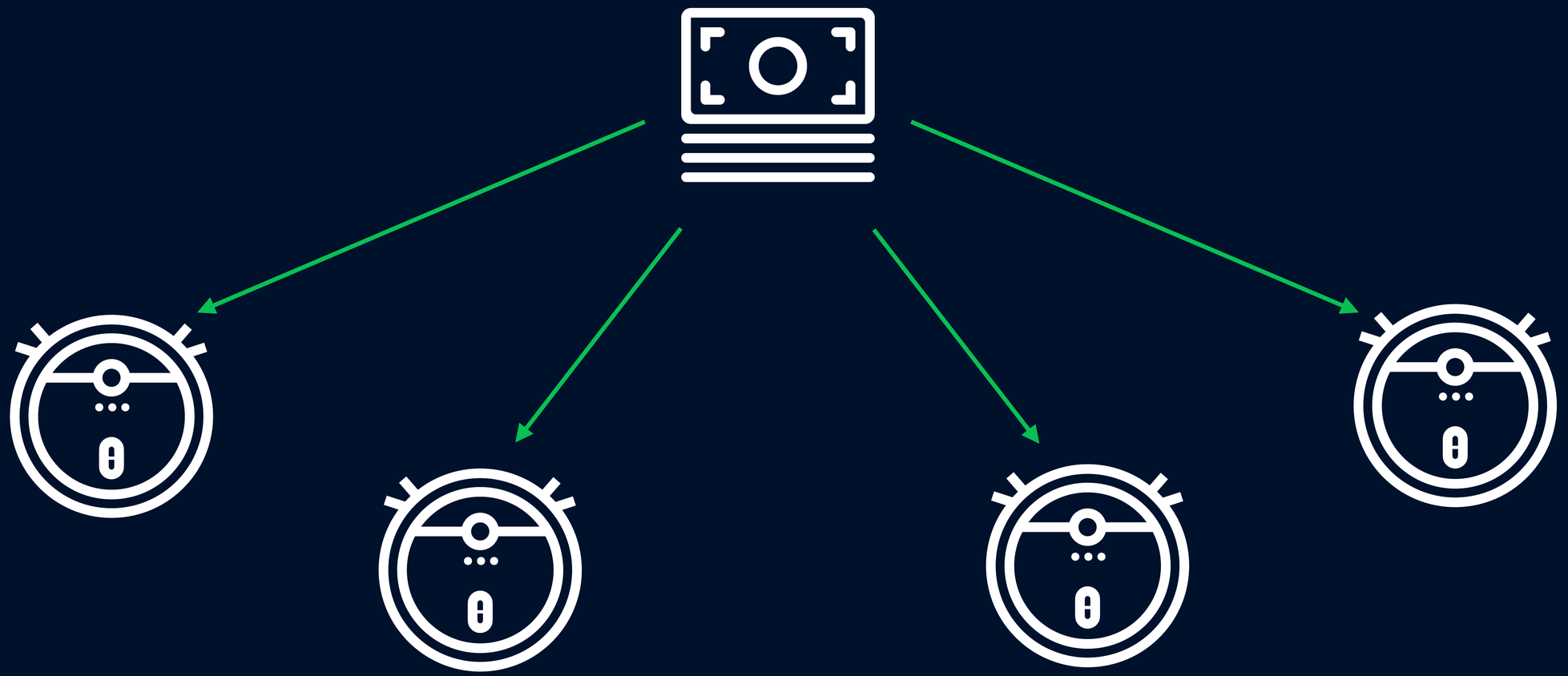
Cost is shared



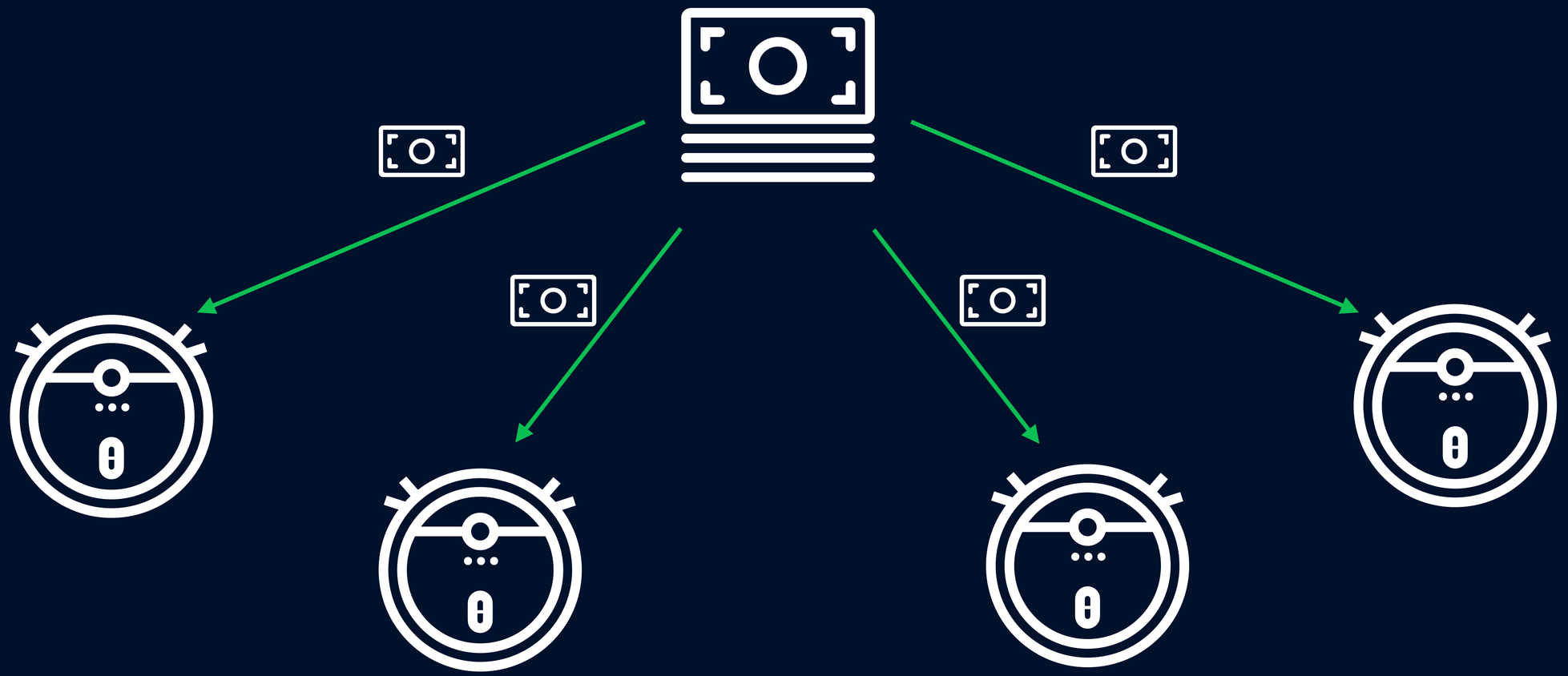
Cost is shared



Cost is shared



Cost is shared





Add workers for more tables in parallel, and increase cost for more speed



Per-table cost settings - outside the shared budget

EDB POSTGRES AI

Autovacuum Tuning Basics



TOMAS VONDRA | JULY 15, 2024

PGConf.dev

11:30–12:20 *Fletcher(1900)*

How Autovacuum Goes Wrong: And Can We Please Make It Stop Doing That?

Robert Haas

PGConf.dev 2024

autovacuum_vacuum_scale_factor



autovacuum_vacuum_threshold



autovacuum_vacuum_cost_limit



autovacuum_vacuum_cost_delay



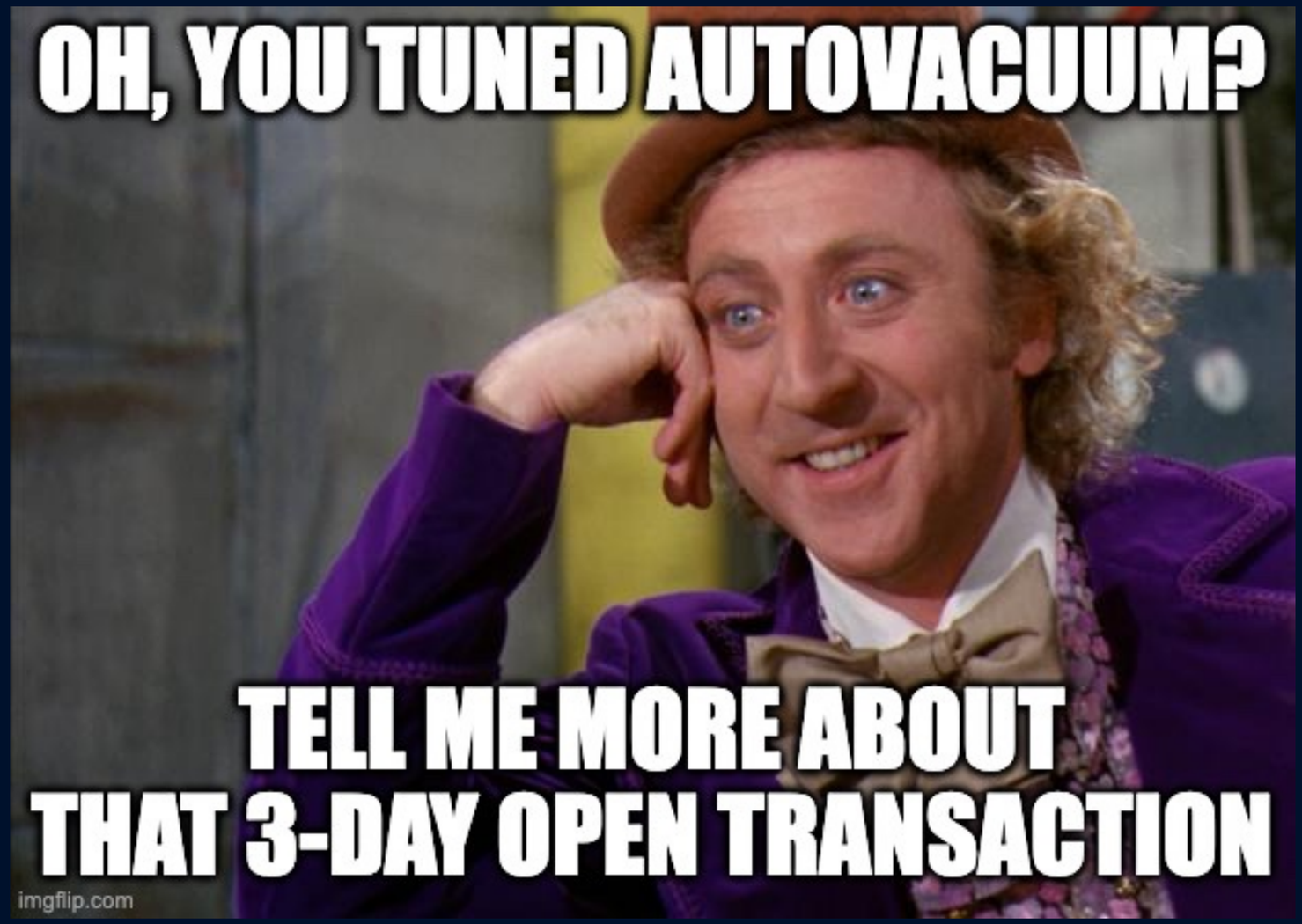
autovacuum_max_workers



When autovacuum cannot run?



en.wikipedia.org/wiki/Valencia By Diego Delso


OH, YOU TUNED AUTOVACUUM?
TELL ME MORE ABOUT
THAT 3-DAY OPEN TRANSACTION
imgflip.com



xmin horizon

postgresql.log

```
2026-10-15 17:14:20.963 UTC [447429] LOG:  automatic vacuum of table
"elcid_meals.public.el_cid_meals": index scans: 0

    pages: 0 removed, 404200 remain, 14200 scanned (3.51% of total), 0 eagerly scanned

    tuples: 0 removed, 9720615 remain, 105000 are dead but not yet removable

    removable cutoff: 42010, which was 113 XIDs old when operation ended

    ...
```

Long running transaction

pg_stat_activity

(backend_xmin)

Long running transaction

`pg_stat_activity`

`(backend_xmin)`

Abandoned replication slot

`pg_replication_slots`

`(xmin)`

Long running transaction

`pg_stat_activity`

`(backend_xmin)`

Abandoned replication slot

`pg_replication_slots`

`(xmin)`

hot_standby_feedback set to on (standby)

`pg_stat_replication (on primary)`

`(backend_xmin)`

Long running transaction

`pg_stat_activity`

`(backend_xmin)`

Abandoned replication slot

`pg_replication_slots`

`(xmin)`

hot_standby_feedback set to on (standby)

`pg_stat_replication (on primary)`

`(backend_xmin)`

Abandoned prepared transaction

`pg_prepared_xacts`

`(transaction)`

AUGUST 12, 2022 • 32 MINUTES

Vacuum

Nikolay and Michael discuss vacuum — what it is, why we need it, some improvements in recent years, and even a creative use for vacuum full.



postgres-ai / postgres-howtos

Code Issues Pull requests Actions Security and quality Insights

main postgres-howtos / 0045_how_to_monitor_xmin_horizon.md Go to file

msdousti Day 45 d45b2c0 · 3 years ago

Preview Code Blame 161 lines (125 loc) · 7 KB Raw

Originally from: tweet, LinkedIn post.

How to monitor xmin horizon to prevent XID/MultiXID wraparound and high bloat

What (auto)vacuum does not fix?



en.wikipedia.org/wiki/Valencia By Rafesmar

True or False?

True or False?

- [ ] Vacuum removes dead tuples

True or False?



Vacuum removes dead tuples

True or False?

- [x] Vacuum removes dead tuples
- [ ] Vacuum frees up space in your tables

True or False?



Vacuum removes dead tuples



Vacuum frees up space in your tables

True or False?

- [x] Vacuum removes dead tuples
- [x]* Vacuum frees up space in your tables
- [ ] Vacuum makes your tables smaller

True or False?



Vacuum removes dead tuples



Vacuum frees up space in your tables



Vacuum makes your tables smaller

`el_cid_meals`
`10M rows`

	HEAP (GB)
Start	3.28

el_cid_meals
10M rows

	HEAP (GB)
Start	3.28
After 8M updates	5.90

`el_cid_meals`
`10M rows`

	HEAP (GB)
Start	3.28
After 8M updates	5.90
After **autovacuum**	5.90

***vacuum_tuncate = off**

`el_cid_meals`
`10M rows`

	HEAP (GB)
Start	3.28
After 8M updates	5.90
After **autovacuum**	5.90
After **VACUUM FULL**	3.28

***vacuum_tuncate = off**

`el_cid_meals`
`10M rows`

	HEAP (GB)
Start	3.28
After 8M updates	5.90
After **autovacuum**	5.90
After **VACUUM FULL**	3.28

	INDEXES (GB)
Start	0.36

***vacuum_tuncate = off**

`el_cid_meals`
`10M rows`

	HEAP (GB)
Start	3.28
After 8M updates	5.90
After **autovacuum**	5.90
After **VACUUM FULL**	3.28

	INDEXES (GB)
Start	0.36
After 8M updates	0.65

***vacuum_tuncate = off**

`el_cid_meals`
`10M rows`

	HEAP (GB)
Start	3.28
After 8M updates	5.90
After **autovacuum**	5.90
After **VACUUM FULL**	3.28

	INDEXES (GB)
Start	0.36
After 8M updates	0.65
After **autovacuum**	0.65

***vacuum_tuncate = off**

`el_cid_meals`
`10M rows`

	HEAP (GB)
Start	3.28
After 8M updates	5.90
After **autovacuum**	5.90
After **VACUUM FULL**	3.28

	INDEXES (GB)
Start	0.36
After 8M updates	0.65
After **autovacuum**	0.65
After **REINDEX CONCURRENTLY**	0.36

***vacuum_tuncate = off**

boringSQL

Home » Posts » VACUUM Is a Lie (About Your Indexes)

VACUUM Is a Lie (About Your Indexes)

2025-12-11 · 10 min · Radim Marek

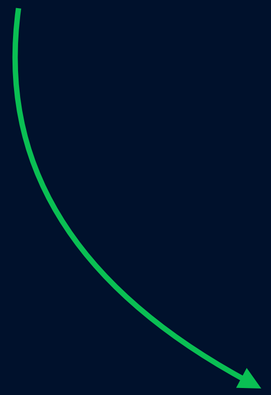



PostgreSQL 19 Updates



en.wikipedia.org/wiki/Valencia By Santi Garcia

REPACK in core

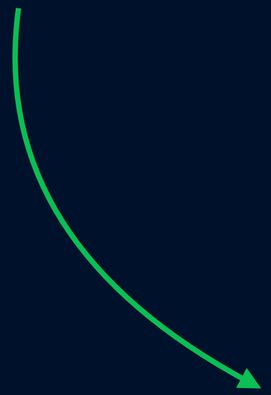




Allow REPACK to rebuild tables without access-exclusive locking

(Antonin Houska, Mihail Nikalayeu, Álvaro Herrera)

Scoring system





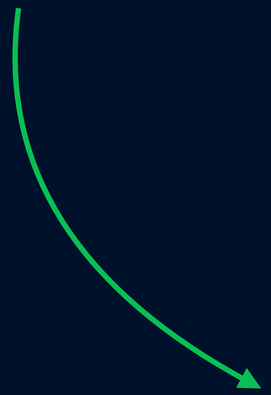
Add scoring system to control the order that tables are processed by autovacuum

(Nathan Bossart)

Add system view `pg_stat_autovacuum_scores` **to report per-table autovacuum details**

(Sami Imseih)

Parallel autovacuum workers



Allow autovacuum to use parallel autovacuum workers

The maximum number of workers is controlled by server variable `autovacuum_max_parallel_workers` and per-table storage parameter `autovacuum_parallel_workers`.

(Daniil Davydov)

Improved observability

Add vacuum initiation details to system view `pg_stat_progress_vacuum`

Add analyze initiation details to system view `pg_stat_progress_analyze`

Add server variable `log_autoanalyze_min_duration` **to log long-running analyze operations by autovacuum operations**

Add reporting of WAL full-page write bytes to VACUUM and ANALYZE logging

(Shinya Kato)

A look at the Elephants Trunk - PostgreSQL 19

Thursday, September 10 at 09:15–10:00



Magnus Hagander

Redpill Linpro

PGDay Lowlands 2026

Table repacking, done right

Thursday, October 23 at 14:55–15:45



Álvaro Herrera

2ndQuadrant — an EDB company



Antonin Houska

CYBERTEC

PGConf.EU 2025

What's New with Monitoring in PostgreSQL 19

Wednesday, October 21 at 09:25–10:15



Gülçin Yıldırım Jelinek

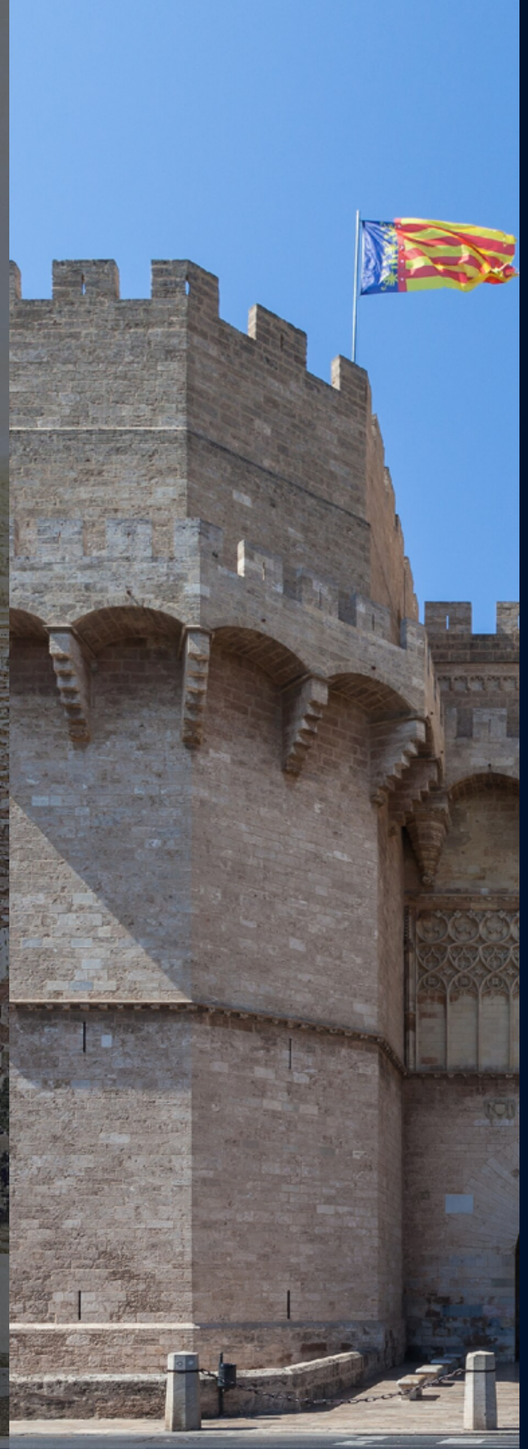
ClickHouse

PGConf.EU 2026

Wrap Up













S DEL DERRIBO DE LAS MURALLAS EN V

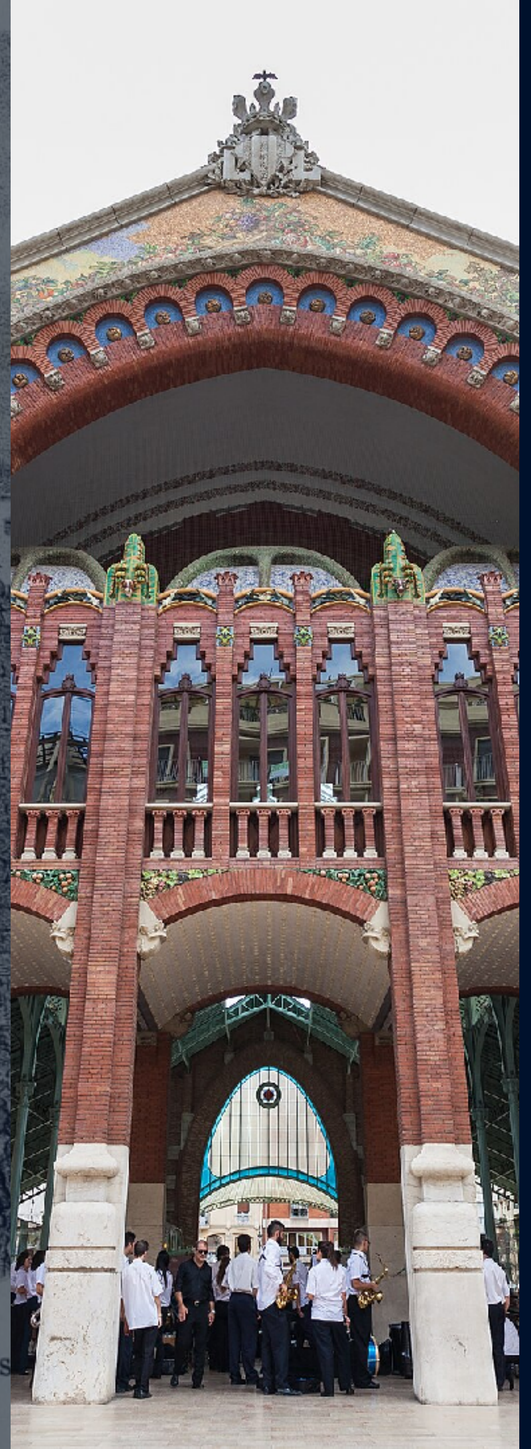
1865, véase la revista de la semana).





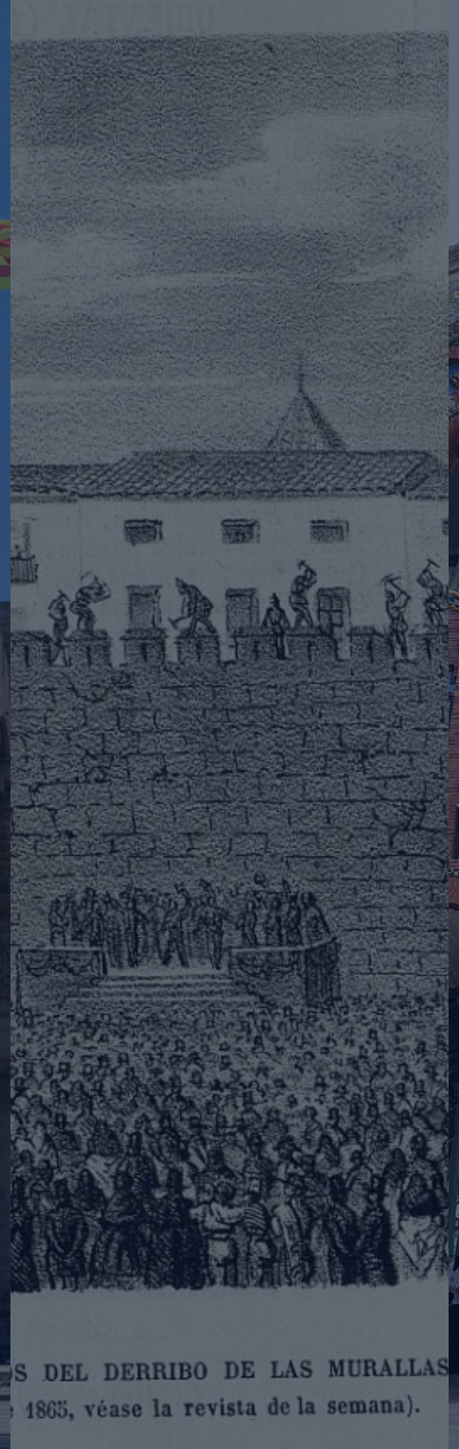


S DEL DERRIBO DE LAS MURALLAS
1865, véase la revista de la semana).









S DEL DERRIBO DE LAS MURALLAS

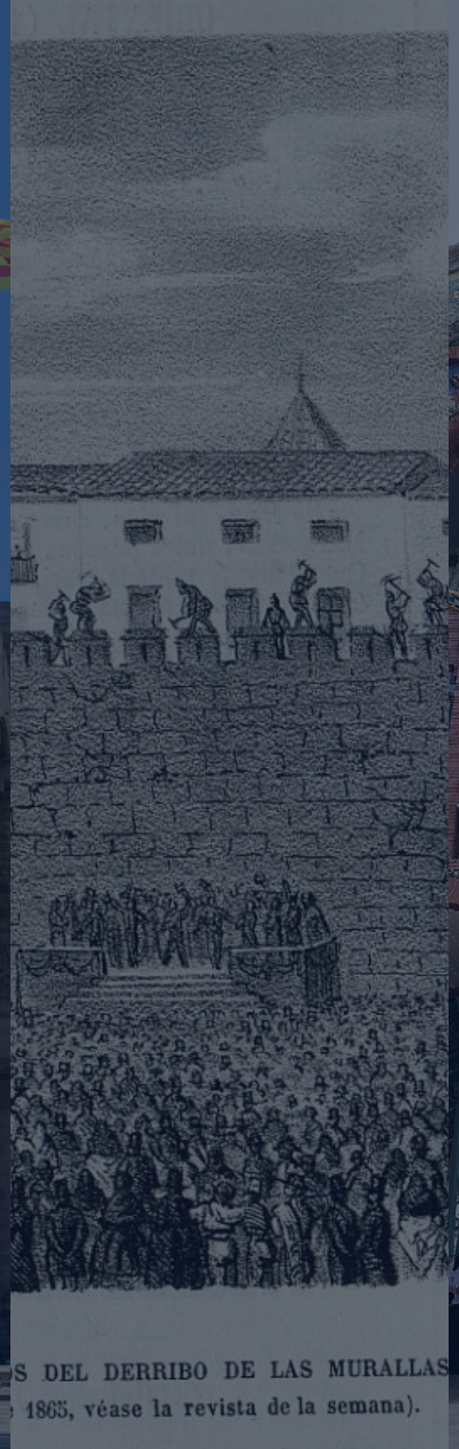
1865, véase la revista de la semana).











S DEL DERRIBO DE LAS MURALLAS
1865, véase la revista de la semana).







“Vacuuming is like exercise.

If it hurts, your are not doing enough of it.”

Robert Haas, Prague 2023

Vacuum is not the problem. Ignoring it is.

Additional resources

Autovacuum Is Running. Is It Actually Working?

adyen

engineered
for ambition